5 FACULTY INFORMATION AND CONTRIBUTIONS (200)

Name of the Faculty Member	PAN No.	University Degree	Date of Receiving Degree	Area of Specialization	Research Paper Publications	Ph.D. Guidance	Faculty received Ph.D. during the assessment year	Current Designation	Date (Designated as Prof./Assoc. Prof.)	Initial Date of Joining	Association Type	At present working with the Institution	In case of No, Date of Leaving
Dr.S.Dharmar	AELPD5780R	M.E./M.Tech. and Ph.D.	21-07-2022	Structural Engineering	7	1	0	Associate Professor	01.01.2023	09.07.2018	Regular	Yes	
Dr.M.Indhumathi	AEEPI4080G	M.E./M.Tech. and Ph.D.	11-01-2021	Structural Engineering	6	4	0	Associate Professor	01.01.2023	23.11.2020	Regular	Yes	
Dr.G.Karthikeyan	BGZPK2732B	M.E./M.Tech. and Ph.D.	27-02-2023	Infrastructure Engineering	6	0	0	Associate Professor	01.02.2024	02.05.2014	Regular	Yes	
Mrs.C.Subha	BWXPS7687H	M.E./M.Tech	28-04-2011	Environmental Management	3	0	0	Assistant Professor		02.06.2014	Regular	Yes	
Mrs.D.Darling Helen Lydia	ALKPL5057D	M.E./M.Tech	16-05-2013	Structural Engineering	0	0	0	Assistant Professor		01.12.2014	Regular	No	12-12-2023
Mr.T.Chockalingam	ATOPC9795P	M.E./M.Tech.	02-12-2015	Structural Engineering	2	0	0	Assistant Professor		02.12.2015	Regular	Yes	
Mrs.R.Kalaimani	COSPK1911G	M.E./M.Tech	19-06-2015	Structural Engineering	1	0	0	Assistant Professor		01.06.2016	Regular	Yes	
Mr.R.Muruganantham	CKKPM5976R	M.E./M.Tech.	25-05-2016	Structural Engineering	2	0	0	Assistant Professor		13.06.2016	Regular	Yes	
Mr.J.Ram Prashath	BQVPR8105H	M.E./M.Tech	25-05-2016	Structural Engineering	0	0	0	Assistant Professor		01.06.2017	Regular	No	23-12-2022
Mr.A.Manicka Mamallan	AZGPM5936Q	M.E./M.Tech	25-05-2017	Structural Engineering	1	0	0	Assistant Professor		01.06.2017	Regular	Yes	
Mrs.A.Leema Margret	ASQPL9580N	M.E./M.Tech.	21-04-2017	Structural Engineering	4	0	0	Assistant Professor		01.10.2020	Regular	Yes	
Mr.V.Ragavan	CDEPR7911L	M.E./M.Tech	25-05-2016	Structural Engineering	4	0	0	Assistant Professor		12.10.2020	Regular	Yes	
Mrs.B.Bharani Baanu	BKTPB2993R	M.E./M.Tech	27-04-2012	Hydrology and Water Resources Engineering	7	0	0	Assistant Professor		01.07.2024	Regular	Yes	
Mr.V.Jeevanantham	BBSPV1999M	M.E./M.Tech	29-06-2015	Geotechnical Engineering	8	0	0	Assistant Professor		08.07.2024	Regular	Yes	

5.1 Student-Faculty Ratio (20)

No. of UG Programs in the Department: 1

Civil Engineering						
Year of Study	**CAY 2024-2025**		**CAYm1 2023-2024**		**CAYm2 2022-2023**	
	Sanction Intake	**Actual admitted through lateral entry students**	**Sanction Intake**	**Actual admitted through lateral entry students**	**Sanction Intake**	**Actual admitted through lateral entry students**
2nd Year	60	3	60	4	60	3
3rd Year	60	4	60	3	60	11
4th Year	60	3	60	11	60	4
Sub Total	**180**	**10**	**180**	**18**	**180**	**18**
Total	**190**		**198**		**198**	
Grand Total	**190**		**198**		**198**	

SFR

Description	**CAY (2024-2025)**	**CAYm1 (2023-2024)**	**CAYm2 (2022-2023)**
Total No. of Students in the Department (S)	**190**	**198**	**198**
No. of Faculty in the Department (F)	**12**	**10**	**11**
Student Faculty Ratio (SFR)	**SFR = S1/F1 = 15.83**	**SFR = S2/F2 = 19.8**	**SFR = S3/F3 = 18.00**
Average SFR	**17.88 ((SFR = SFR1+SFR2+SFR3)/3)**		
F=Total Number of Faculty Members in the Department			

5.1.1 Provide the information about the regular and contractual faculty as per the format mentioned below:

	Total number of regular faculty in the department	Total number of contractual faculty in the department
CAY (2024-2025)	12	0
CAYm1 (2023-2024)	10	0
CAYm2 (2022-2023)	11	0

Average SFR for three assessment years: 17.88

5.2. Faculty Cadre Proportion (25)

The reference Faculty cadre proportion is 1(F1):2(F2):6(F3)

F1: Number of Professors required = 1/9 x Number of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (N) as per 5.1

F2: Number of Associate Professors required = 2/9 x Number of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (N) as per 5.1

F3: Number of Assistant Professors required = 6/9 x Number of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (N) as per 5.1

Year	Professors		Associate Professors		Assistant Professors	
	Required F1	Available	Required F2	Available	Required F3	Available
CAY (2024-25)	1	0	2	3	6	9
CAY m1(2023-24)	1	0	2	2	6	8
CAYm2 (2022-23)	1	0	2	0	6	11
Average Numbers	RF1=1	AF1=0	RF2=2	AF2=1.67	RF3=6	AF3= 9.33
Cadre Ratio Marks						**14.04**

5.3. Faculty Qualification (25)

FQ =2.5 x [(10X +4Y)/F)] where x is no. of regular faculty with Ph.D., Y is no. of regular faculty with M.Tech. F is no. of regular faculty required to comply 20:1 Faculty Student ratio (no. of faculty and no. of students required are to be calculated as per 5.1)

Years	X	Y	F	FQ=2.5 x [(10X +4Y)/F)]
CAY (2024-25)	**3**	9	9	18.33
CAYM1 (2023-24)	**3**	7	9	16.11
CAYM2 (2022-23)	**2**	9	9	15.56
Average Assessment				**16.67**

5.4. Faculty Retention (25)

No. of regular faculty members in CAYm1 = 10 CAY = 12
CAYm2 = 11

Description	2023-2024	2024-2025
No of Faculty Retained	10	10
Total No of Faculty	10	12
% of Faculty Retained	100	83.33

5.5. Innovations by the Faculty in Teaching and Learning (20)

Innovations by the Faculty in Teaching and Learning:

The teaching and learning process is highly valued by the Department of Civil Engineering faculty. The faculty members employ various cutting-edge pedagogical strategies in their teaching and learning process in addition to traditional teaching techniques like chalk & talk and power point presentations to boost the students' understanding. Among the cutting-edge pedagogical methods frequently employed by the faculty of civil engineering are include:

- Team quizzes
- Flipped classroom
- Mind maps
- LMS based learning
- One-minute paper
- Model based learning
- Industrial & Field visit
- Video lectures
- Group discussions
- Reflective Journal
- Experiential learning
- Handouts
- Think Pair Share
- Collaborative learning

The innovative pedagogic teaching techniques implemented by the faculty of Civil Engineering are available for review by academicians and public in the form of documents available in the course files and Learning Management System (LMS). The faculty members issue handouts and scaffolding materials to students extensively for their courses in order to improve the understanding of the students.

Think pair share

Think-Pair-Share is an instructional strategy that encourages student engagement, collaboration, and critical thinking. It is widely used in classrooms to enhance learning by allowing students to think independently, discuss their ideas with a partner, and then share their thoughts with class. Think-Pair-Share activities often lead to deeper engagement with content, better communication skills, and enhanced social learning. This activity is conducted for the course strength of materials II. As a result, the class attained 100% pass percentage in the respective end semester exam conducted.

Fig. 5.1 Think pair share while handling Strength of Materials-II

Collaborative learning

Collaborative Learning is an educational approach that emphasizes the importance of teamwork and collective problem-solving among students. It involves students working together in groups to achieve shared learning goals. This method not only enhances academic achievement but also develops essential social skills, critical thinking, and a sense of community within the classroom. This activity is conducted for the course strength of materials I. As a result, the class pass percentage improved to 72.73% in the respective end-semester exam.

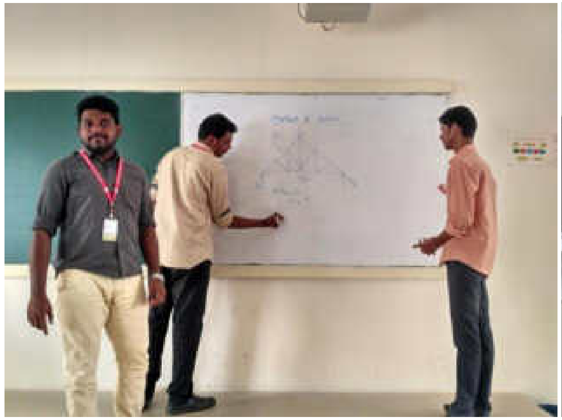




Fig. 5.2 Collaborative Learning Strength of Materials I

Experiential Learning

Experiential learning is a process through which individuals gain knowledge, skills, and competencies by engaging in direct experience and reflection. Students enhance their critical thinking, problem-solving, and practical skills, while also gaining self-awareness and emotional intelligence. This activity was conducted for the course Applied Hydraulics Engineering. As a result, the class pass percentage increased significantly, from 52.38% in the previous year to 82.22% in the respective end-semester exam

Fig. 5.3 Experiential Learning Applied Hydraulics Engineering

The following table provides the details of the innovative pedagogic techniques implemented by the faculty members of Department of Civil Engineering in their courses. (Table 5.1)

Table 5.1 Innovative Teaching Learning Methodologies Adopted

Name of the Faculty member	Innovative pedagogic techniques adopted	Course Code & Course Name	Semester	Academic Year	Availability
Mrs.B.Bharani Baanu	Theory to Practical	CE3301 Fluid Mechanics	III	2024-2025	In Course File, College website
Mrs.B.Bharani Baanu	Video Lecture	CE3301 Fluid Mechanics	III	2024-2025	In Course File, College website
Mrs.B.Bharani Baanu	Video Lecture	AI3404 Hydrology and Water Resources Engineering	VII	2024-2025	In Course File, College website
Dr.G.Karthikeyan	Experimental Learning	CE3403 Concrete Technology	IV	2023-2024	In Course File, College website
Dr.M.Indhumathi	Simulation Tool	CE3502 Structural Analysis I	V	2023-2024	In Course File, College website
Mrs.R. Kalaimani	Mind Maps	CCE333 Environmental Impact Assessment	VI	2023-2024	In Course File, College website
Mrs.R. Kalaimani	Virtual Laboratory	CE3401 Applied Hydraulic Engineering	IV	2023-2024	In Course File, College website
Mrs.R. Kalaimani	Theory to Practical	CE3401 Applied Hydraulic Engineering	IV	2023-2024	In Course File, College website
Mr.V.Ragavan	Theory to Practical	CE3404 Soil Mechanics	IV	2023-2024	In Course File, College website
Dr.M.Indhumathi	Simulation Tool	CE3602 Structural Analysis II	VI	2023-2024	In Course File, College website
Mrs.A.Leema Margret	Quick Write	CE8702 Railways, Airports, Docks and Harbour Engineering	VII	2023-2024	In Course File, College website
Mrs.A.Leema Margret	Crossword Puzzle	CE8702 Railways, Airports, Docks and Harbour Engineering	VII	2023-2024	In Course File, College website
Mr.V.Ragavan	Mind Map	CCE331 Air and Noise Pollution Control Engineering	V	2023-2024	In Course File, College website
Mr.V.Ragavan	Four Corner Four Questions	CCE331 Air and Noise Pollution Control Engineering	V	2023-2024	In Course File, College website
Mrs.R. Kalaimani	Virtual Laboratory	CE3401 Applied Hydraulic Engineering	IV	2022-2023	In Course File, College website

Mrs.R. Kalaimani	Theory to Practical	CE3401 Applied Hydraulic Engineering	IV	2022-2023	In Course File, College website
Mr.G. Karthikeyan	Experimental learning	CE3403 Concrete Technology	IV	2022-2023	In Course File, College website
Mr.G. Karthikeyan	Open Book Test	CE3403 Concrete Technology	IV	2022-2023	In Course File, College website
Mr.V. Ragavan	One minute paper	CE3402 Strength of Materials	IV	2022-2023	In Course File, College website
Mr.V. Ragavan	Theory to Practical	CE3402 Strength of Materials	IV	2022-2023	In Course File, College website
Mrs. A. Leema Margret	Theory to Practice	CE8603 Irrigation Engineering	IV	2022-2023	In Course File, College website
Mrs. A. Leema Margret	Visible Quiz	CE8603 Irrigation Engineering	VI	2022-2023	In Course File, College website
Mrs. A. Leema Margret	One minute paper	CE8603& Irrigation Engineering	VI	2022-2023	In Course File, College website
Dr.M. Indhumathi	Simulation Tool	CE8602 Structural Analysis II	VI	2022-2023	In Course File, College website
Mrs.D. Darling Helen Lydia	Virtual Laboratory	CE3404 Soil Mechanics	VI	2022-2023	In Course File, College website
Mrs.R.Kalaimani	Virtual Laboratory	CE3301 Fluid Mechanics	III	2022-2023	In Course File, College website
Mrs.R.Kalaimani	Theory to Practice	CE3301Fluid Mechanics	III	2022-2023	In Course File, College website
Mrs.R.Kalaimani	One Minute Paper	CE3301 Fluid Mechanics	III	2022-2023	In Course File, College website
Mrs.D.Darling Helen Lydia	Visible Quiz	CE3302-Construction Materials& Technology	III	2022-2023	In Course File, College website
Mrs.C.Subha	Visible Quiz	EN8491-Water Supply Engineering	III	2022-2023	In Course File, College website
Mrs.C.Subha	One Minute Paper	EN8491-Water Supply Engineering	III	2022-2023	In Course File, College website

Mrs.C.Subha	Reflective Journal	EN8491-Water Supply Engineering	III	2022-2023	In Course File, College website
Mrs.C.Subha	Visible Quiz	CE3303-Water Supply& Waste water Engineering	V	2022-2023	In Course File, College website
Mrs.C.Subha	Reflective Journal	CE3303-Water Supply& Waste water Engineering	V	2022-2023	In Course File, College website
Mrs.C.Subha	Virtual Laboratory	CE3303-Water Supply& Waste water Engineering	V	2022-2023	In Course File, College website
Mr.G.Karthikeyan	Open Book Test	CE8701-Estimation,Costing & Valuation Engineering	VII	2022-2023	In Course File, College website
Dr.M.Indhumathi	Simulation Tool	CE8502& Structural Analysis I	V	2022-2023	In Course File, College website
Mr.V. Ragavan	Visible Quiz	OCE551-Airpollution and Control Engineering	V	2022-2023	In Course File, College website
Mr.V. Ragavan	Four Corner Four questions	OCE551-Airpollution and Control Engineering	V	2022-2023	In Course File, College website
Mr.V. Ragavan	One minute paper	OCE551-Airpollution and Control Engineering	V	2022-2023	In Course File, College website
Mrs.R.Kalaimani & Mrs.A.Leema Margret	One minute paper	EN8591-Municipal Solid Waste Management	VII	2022-2023	In Course File, College website
Mrs.R.Kalaimani & Mrs.A.Leema Margret	Reflective Journal	EN8591-Municipal Solid Waste Management	VII	2022-2023	In Course File, College website
Mrs.R.Kalaimani & Mrs.A.Leema Margret	Visible Quiz	EN8591-Municipal Solid Waste Management	VII	2022-2023	In Course File, College website
Mrs.A.Leema Margret	Cross Word and Jig Jaw Puzzle	GE8071-Disaster Management	VII	2022-2023	In Course File, College website
Mrs.A.Leema Margret	Disaster Supplies kit Concentration	GE8071-Disaster Management	VII	2022-2023	In Course File, College website
Mrs.A.Leema Margret	Visible Quiz	GE8071-Disaster Management	VII	2022-2023	In Course File, College website
Mrs.R.Kalaimani	Virtual Laboratory	CE8403 Applied Hydraulic Engineering	IV	2021-2022	In Course File, College website

Mrs.R.Kalaimani	Theory to Practice	CE8403 Applied Hydraulic Engineering	IV	2021-2022	In Course File, College website
Mr.G. Karthikeyan	Experimental learning	CE8404 Concrete Technology	IV	2021-2022	In Course File, College website
Mr.V.Ragavan	One minute paper	CE8005Air pollution and Control Engineering	VI	2021-2022	In Course File, College website
Mr.V.Ragavan	Four Corner Four Question	CE8005Air pollution and Control Engineering	VI	2021-2022	In Course File, College website
Mrs.C. Subha	Virtual Lab	EN8592 Waste Water Engineering	VI	2021-2022	In Course File, College website
Mrs.C. Subha	Crossword Puzzle	EN8592 Waste Water Engineering	VI	2021-2022	In Course File, College website
Mrs. A. Leema Margret	Visible Quiz	CE8603 Irrigation Engineering	VI	2021-2022	In Course File, College website
Mrs. A. Leema Margret	One minute paper	CE8603 Irrigation Engineering	VI	2021-2022	In Course File, College website
Dr.M.Indhumathi	Simulation Tool	CE8602 Structural Analysis II	VI	2021-2022	In Course File, College website
Mr.J.Ram Prashath	Cross Word Puzzle	CE8401 Construction Techniques and Practices	IV	2021-2022	In Course File, College website
Mr.J.Ram Prashath	One Minute Paper	CE8401 Construction Techniques and Practices	IV	2021-2022	In Course File, College website
Mrs.D.Darling Helen Lydia	Virtual Laboratory	CE8491 Soil Mechanics	IV	2021-2022	In Course File, College website
Mrs.D.Darling Helen Lydia	Laboratory Visit	CE8491 Soil Mechanics	IV	2021-2022	In Course File, College website
Mrs.R. Kalaimani	Theory to Practice	CE8302 Fluid Mechanics	III	2021-2022	In Course File, College website
Mrs.R. Kalaimani	One minute paper	CE8361 Surveying Laboratory	III	2021-2022	In Course File, College website
Mrs.R. Kalaimani	Mind Map	CE8591 Foundation Engineering	V	2021-2022	In Course File, College website

Dr.M.Indhumathi	Simulation Tool	CE8502 Structural Analysis I	V	2021-2022	In Course File, College website
Dr.M.Indhumathi	Quiz	CE8391 Construction Materials	V	2021-2022	In Course File, College website
Mr.J.Ramprashath	Cross Word Puzzle	CE8392 Engineering Geology	III	2021-2022	In Course File, College website
Mr.J.Ramprashath	One minute paper	CE8392 Engineering Geology	III	2021-2022	In Course File, College website
Mr.J.Ramprashath	Theory to Practice	CE8392 Engineering Geology	III	2021-2022	In Course File, College website
Mr.V. Ragavan	One minute paper	CE8301Strength of Materials I	III	2021-2022	In Course File, College website
Mr.V. Ragavan	Virtual Lab	CE8301Strength of Materials I	III	2021-2022	In Course File, College website
Mrs.A.Leema Margret	One minute paper	EN8591 Municipal Solid Waste Management	VII	2021-2022	In Course File, College website
Mrs.A.Leema Margret	Recycle Relay	EN8591 Municipal Solid Waste Management	VII	2021-2022	In Course File, College website
Mrs.A.Leema Margret	Cross Word and Jig Jaw Puzzle	GE8071 Disaster Management	V	2021-2022	In Course File, College website
Mrs.A.Leema Margret	Reflective Journal	GE8071 Disaster Management	V	2021-2022	In Course File, College website
Mrs.A.Leema Margret	Visible Quiz	GE8071 Disaster Management	V	2021-2022	In Course File, College website
Mr.V.Jeevanantham	Google Site (Mindmap)	CE3503 Foundation Engineering	V	2024-2025	College website

Other highlights of the teaching learning process in the department are:

- The department classrooms are equipped with modern teaching aids like LCD projectors; Internet enabled computer systems and high quality audio system.
- Apart from the college main library, the civil engineering department has a separate library to cater to the needs of the civil engineering students and faculty.
- The department library is easy and open accessible to all the civil engineering students and faculty.
- The faculties have been participating/presenting papers in national/international conferences and publish their articles in national/international journals to enrich their knowledge.
- The research laboratories established by the department like RIT-Centre of Excellence for Building Information Modelling (BIM), RIT ICT Academy Centre of Excellence for Design Powered by Autodesk, Centre for Geospatial Technology, Construction Practices Laboratory and Project & Consultancy Laboratory in the development of faculty related to delivering lectures in advanced topics.
- The laboratory courses are handled by the faculty with utmost care in such a way to ensure the enrichment of the practical skill and knowledge of the students in the respective laboratory experiments.
- The respective faculty members handling the laboratory courses prepare an extensive Laboratory Manual which helps the students to understand and practice the applications of the theoretical concepts taught in the classrooms.

The department and institute make great efforts to teach the faculty in new pedagogic practises by allowing the faculty members to participate in numerous workshops, FDPs, STTPs and online courses relevant to pedagogy organized at many esteemed institutions. In order to improve their teaching skills, the faculty members participate in a variety of programmes. The details of the pedagogy-related workshops, FDPs, STTPs and online courses that the faculty of civil engineering attended are listed in table 5.5.6.1.

Table 5.2 Workshops/FDPs/STTPs/Online course related to Pedagogy attended by Faculty

S. No.	Name of the Faculty	Type of Program	Title of Program	Organized by	Program Period	Academic Year
1.	Dr.G.Karthikeyan	Online Course	Accreditation and Outcome Based Learning	IIT Kharagpur	8 Weeks	2024-2025
			Module 5: Technology Enabled Learning & Life Long Self Learning	NITTTR, Chennai	July – August 2022	2022-2023
			Module 6: Student Assessment and Evaluation	NITTTR, Chennai	July – August 2022	2022-2023
			Module 4 - Instructional Planning and Delivery	NITTTR, Bhopal	September - October 2021	2021-2022
2.	Mr.T.Chockalingam	Faculty Competence Enhancement Programme	LPB01 – Reverse Engineering	LEAP	2 days	2023-2024
		Online Course	Module 4 - Instructional Planning and Delivery	NITTTR, Bhopal	September - October 2021	2021-2022
			Module 5: Technology Enabled Learning & Life Long Self Learning	NITTTR, Chennai	February - March 2022	2021-2022
			Module 6: Student Assessment and Evaluation	NITTTR, Chennai	February - March 2022	2021-2022
3.	Mr.R.Muruganantham	Online Course	Module 5: Technology Enabled Learning & Life Long Self Learning	NITTTR, Chennai	July – August 2022	2022-2023
			Module 4 - Instructional Planning and Delivery	NITTTR, Bhopal	September – October 2021	2021-2022
			Module 6: Student Assessment and Evaluation	NITTTR, Chennai	February – March 2022	2021-2022

			SWAYAM	Module 2 - Professional Ethics & Sustainability	-	February 2023	2022-2023
			SWAYAM	Module 8 – Institutional Management and Administrative Procedures	-	February 2023	2022-2023
4.	Mr.A.Manicka Mamallan	Online Course	Module 4 - Instructional Planning and Delivery	NITTTR, Bhopal	September – October 2021	2021-2022	
5.	Mrs.A.Leema Margret	Faculty Competence Enhancement Programme	LPB01 – Reverse Engineering	LEAP	2 days	2023-2024	
		Online Course	Module 1: Orientation towards Technical Education and Curriculum Aspects	NITTTR, Chennai	September - October 2023	2023-2024	
			Module 6: Student Assessment and Evaluation	NITTTR, Chennai	February - March 2023	2022-2023	
			Module 3 - Communication Skills Modes & Knowledge Dissemination	NITTTR, Chandigarh	July - August 2022	2021-2022	
			Module 4 - Instructional Planning and Delivery	NITTTR, Bhopal	July - August 2022	2021-2022	
6.	Mr.V.Ragavan	Online Course	Module 6: Student Assessment and Evaluation	NITTTR, Chennai	February - March 2023	2022-2023	
			Module 3 - Communication Skills Modes & Knowledge Dissemination	NITTTR, Chandigarh	July - August 2022	2021-2022	
			Module 4 - Instructional Planning and Delivery	NITTTR, Bhopal	July - August 2022	2021-2022	
7.	Mrs.D.Darling Helen Lydia	Online Course	IUCEE International Engineering Educator's Certification Program	IUCEE		2021-2022	

8.	Mr.J.Ram Prashath	Online Course	Module 4 - Instructional Planning and Delivery	NITTTR, Bhopal	July - August 2022	2021-2022

The department and institute encourage faculty members to enhance their teaching skill and technical knowledge by attending and completing various online courses and webinars on advanced topics and pedagogic techniques through various platforms such as NPTEL, SWAYAM, Coursera & EDX etc. in order to keep pace with the advanced level of knowledge and skills. The following table provides the details of online courses completed the faculty of Civil Engineering.

Table 5.3 Online Courses completed by faculty for Technical Competency

S. No.	Name of the Faculty Member	Organizing Institute	Platform	Title of Online Course	Duration of Course	Period of Course	Academic Year
1.	Dr.S.Dharmar	University of Copenhagen	Coursera	Sustainable Tourism – promoting environmental public health	-	30.12.2022	2022-2023
		Indian Green Building Council	IGBC	Green Buildings and Built Environment	2 Weeks	13^{th} – 24^{th} March 2023	2022-2023
2.	Dr.M.Indhumathi	IIT Madras	SWAYAM	Admixtures and special concretes	12 Weeks	Jul-Oct 2023	2023-2024
		IOV Registered Valuers Foundation	-	Land & Building	10 Days	21.10.2022 to 30.10.2022	2022-2023
3.	Dr.G.Karthikeyan	IIT Kharagpur	SWAYAM	Accreditation and Outcome Based Learning	8 Weeks	Aug-Oct 2024	2024-2025
		IIT Madras	SWAYAM	Admixtures and special concretes	12 Weeks	Jul-Oct 2023	2023-2024
		Duke University	Coursera	Data Science Math Skills	-	Feb 9, 2024	2023-2024

S. No.	Name of the Faculty Member	Organizing Institute	Platform	Title of Online Course	Duration of Course	Period of Course	Academic Year
		IIT Madras	SWAYAM	Maintenance and Repair of Concrete Structures	12 Weeks	Jan-Apr 2024	2023-2024
		Hong Kong University of Science and Technology	Coursera	Matrix Algebra for Engineers / Completed	-	May 27, 2023	2022-2023
		IIT Madras	SWAYAM	Concrete Technology	12 Weeks	Jan- April 2023	2022-2023
4.	Mrs.C.Subha	IIT Bombay	SWAYAM	Remote Sensing: Principles and Applications	12 Weeks	Jul-Oct 2024	2024-2025
		IIT Guwahati	SWAYAM	Municipal Solid Waste Management	12 Weeks	Jul-Oct 2023	2023-2024
		EIT Digital	Coursera	Mastering Digital Twins	-	Sep 8, 2023	2023-2024
		Duke University	Coursera	Data Science Math Skills	-	Mar 13, 2024	2023-2024
		IIT- Roorkee	SWAYAM	Geographic Information Systems	12 Weeks	Jan-Apr 2024	2023-2024
		University of Copenhagen	Coursera	Sustainable Tourism – promoting environmental public health	-	Dec 2022	2022-2023
		Hong Kong University of Science and Technology	Coursera	Matrix Algebra for Engineers	-	Apr 2023	2022-2023
5.	Mr.T.Chockalingam	IIT Guwahati	SWAYAM	Fundamentals of Additive Manufacturing Technologies	12 Weeks	July – October 2024	2024-2025
		IIT Kanpur	SWAYAM	Metal Additive Manufacturing	12 Weeks	July – October 2024	2024-2025
		NITTTR Bhopal	SWAYAM	Module 2 - Professional Ethics & Sustainability	-	February – March 2023	2023-2024

S. No.	Name of the Faculty Member	Organizing Institute	Platform	Title of Online Course	Duration of Course	Period of Course	Academic Year
		IIT Madras	SWAYAM	Concrete Technology	12 Weeks	Jan- April 2023	2023-2024
6.	Mrs.R.Kalaimani	IIT Madras	SWAYAM	Building Materials as a Cornerstone to Sustainability	12 Weeks	July – October 2024	2024-2025
		Ecole Polytechnique	Coursera	Fundamentals of Fluid-Solid Interactions	-	August 2024	2024-2025
		NITTTR	SWAYAM	Module 2 Professional Ethics & Sustainability	-	September 2023	2023-2024
		NITTTR	SWAYAM	Module 8 Institutional Management & Administrative Procedures	-	September 2023	2023-2024
		Duke University	Coursera	Data Science Math Skills	-	Mar 7, 2024	2023-2024
		IIT Madras	SWAYAM	Concrete Technology	12 Weeks	Jan-Apr 2024	2023-2024
		University of Copenhagen	Coursera	Sustainable Tourism – promoting environmental public health	-	Jan 2023	2023-2024
		Hong Kong University of Science and Technology	Coursera	Matrix Algebra for Engineers / Completed	-	Apr2023	2023-2024
7.	Mr.R.Muruganantham	Duke University	Coursera	Data Science Math Skills	-	Feb 2024	2023-2024
		NITTTR	SWAYAM	Module 2 - Professional Ethics & Sustainability	-	February 2023	2022-2023
		NITTTR	SWAYAM	Module 8 – Institutional Management and Administrative Procedures	-	February 2023	2022-2023
		Hong Kong University of	Coursera	Matrix Algebra for Engineers	-	August 28,2023	2023-2024

S. No.	Name of the Faculty Member	Organizing Institute	Platform	Title of Online Course	Duration of Course	Period of Course	Academic Year
		Science and Technology					
8.	Mr.A.Manicka Mamallan	IIT- Roorkee	SWAYAM	GPS Surveying	4 Weeks	July – August 2022	2022-2023
		National Institute of Disaster Management		"Understanding Recurring Heatwaves Risk Impact and the Way Forward for Resilience"	3 Days	26 Jul 2022 to 28 Jul 2022	2022-2023
9.	Mrs.A.Leema Margret	IIT Madras	SWAYAM	Building Materials as a Cornerstone to Sustainability	12 Weeks	July – October 2024	2024-2025
		Duke University	Coursera	Data Science Math Skills	-	Mar 2024	2023-2024
		National Institute of Disaster Management	-	"Understanding Recurring Heatwaves Risk Impact and the Way Forward for Resilience"	3 Days	26 Jul 2022 to 28 Jul 2022	2022-2023
		Hong Kong University of Science and Technology	Coursera	Matrix Algebra for Engineers	-	May 27, 2023	2022-2023
		Indian Green Building Council	IGBC	Green Buildings and Built Environment	12 Days	13 March – 24 March 2024	2023-2024
10.	Mr.V.Ragavan	IIT Madras	SWAYAM	Building Materials as a Cornerstone to Sustainability	12 Weeks	July – October 2024	2024-2025
		IIT Madras	SWAYAM	Admixtures and special concretes	12 Weeks	Jul-Oct 2023	2023-2024
		Duke University	Coursera	Data Science Math Skills	-	Mar 2024	2023-2024
		IIT Madras	SWAYAM	Concrete Technology	12 Weeks	Jan-Apr 2024	2023-2024

S. No.	Name of the Faculty Member	Organizing Institute	Platform	Title of Online Course	Duration of Course	Period of Course	Academic Year
11.	Mrs.B.Bharani Baanu	IIT Guwahati	SWAYAM	Natural Resource Management	12 Weeks	July – August 2024	2024-2025
12.	Mr.V.Jeevanantham	IIT Bombay	SWAYAM	Geotechnical Engineering Laboratory	4 Weeks	July – August 2024	2024-2025
		IIT Kharagpur	SWAYAM	Ground Improvement	12 Weeks	July – October 2024	2024-2025
13.	Mrs.D.Darling Helen Lydia	University of Copenhagen	Coursera	Sustainable Tourism – promoting environmental public health	-	Jul 2023	2023-2024
		IIT - Kanpur	SWAYAM	Geology and Soil Mechanics	12 Weeks	Jan- April 2023	2023-2024
		Hong Kong University of Science and Technology	Coursera	Matrix Algebra for Engineers	-	May 22, 2023	2023-2024

Study materials uploaded in the public platforms by the faculty

Study materials in the form of PPT/PDF are uploaded in the platform by the faculty such as SlideShare, Google sites. Moreover, lecture videos are uploaded in YouTube for benefit of students and the general public for peer review and critique. Samples with access links of the SlideShare, Google sites and YouTube are shown below.

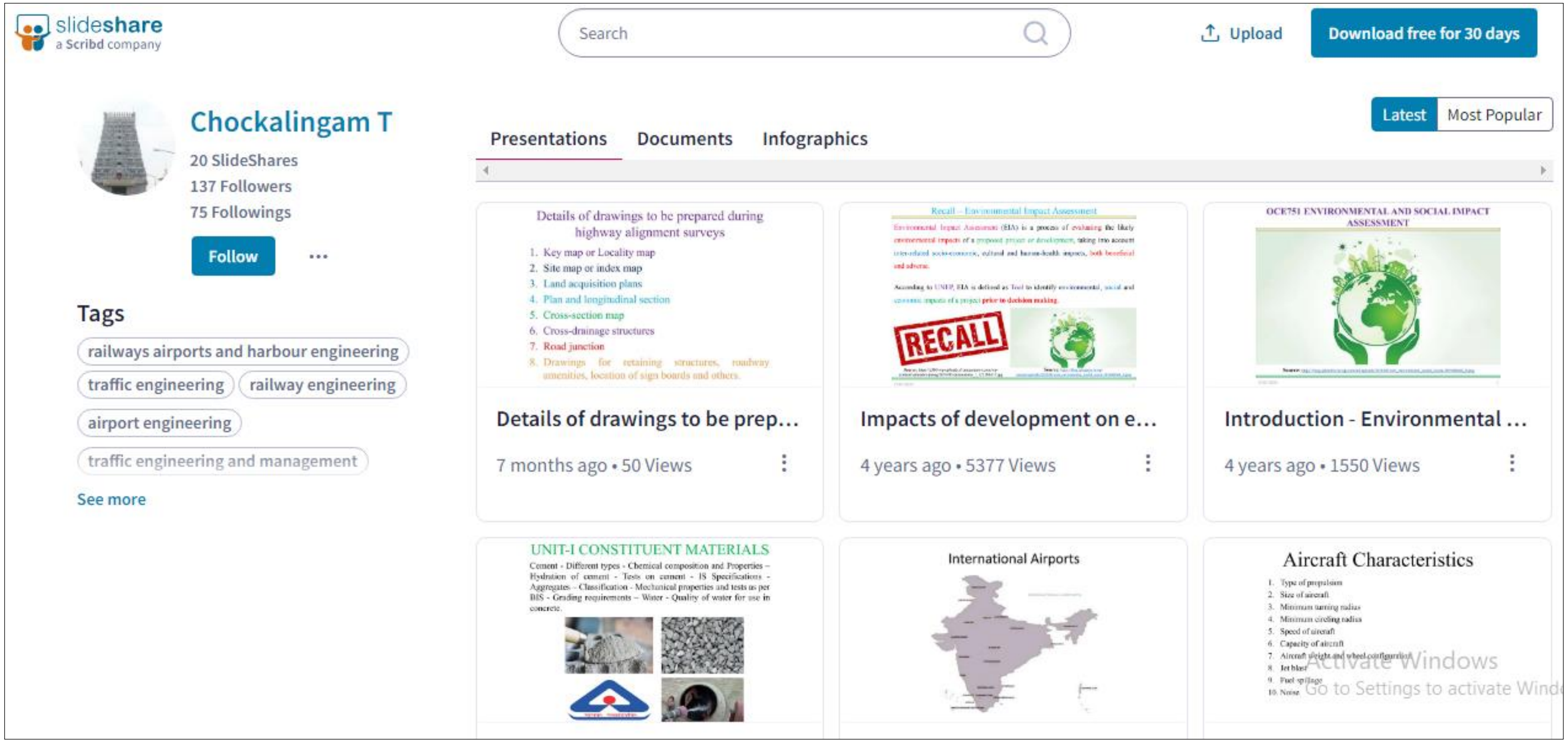
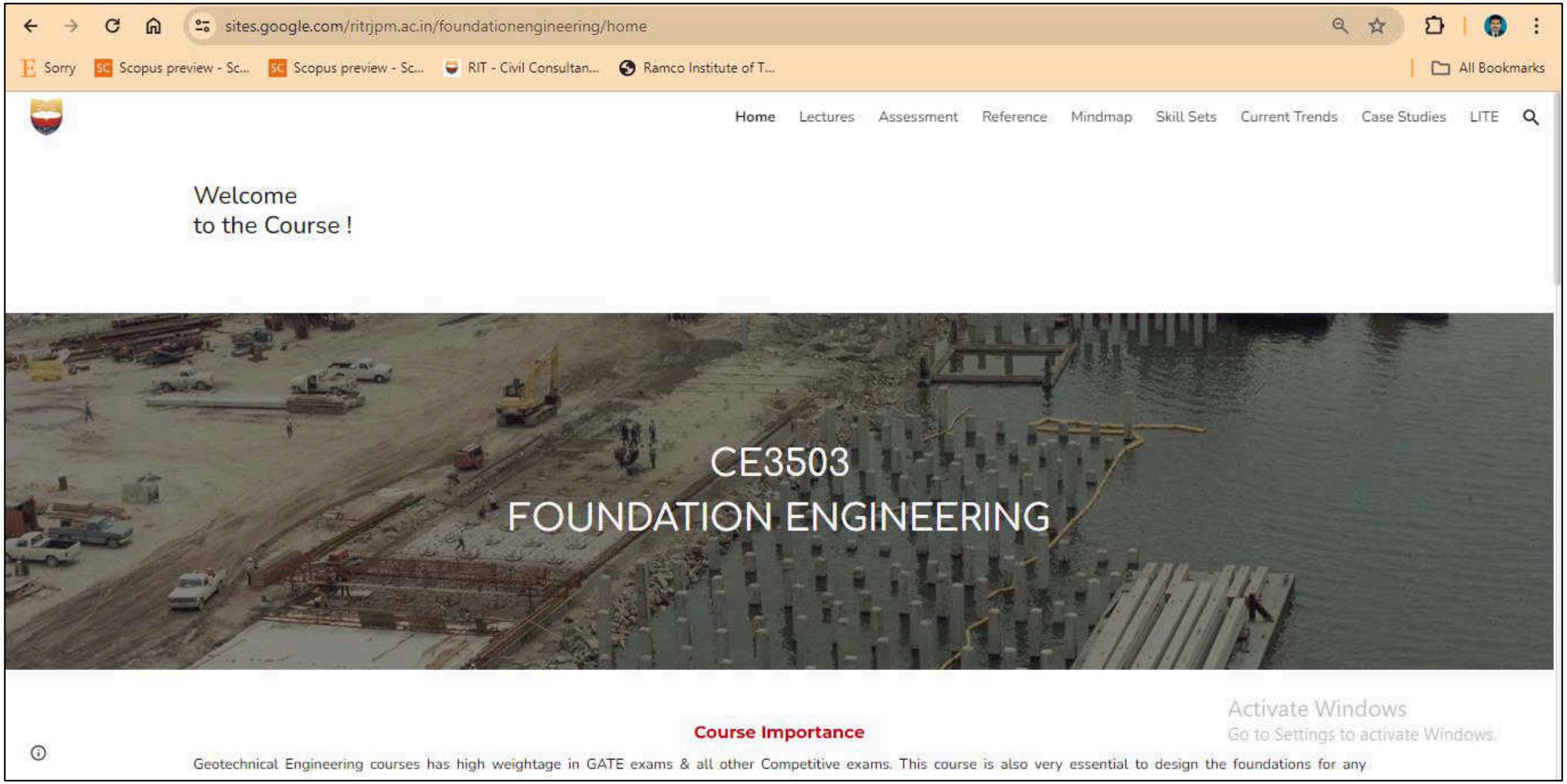


Fig. 5.4 Sample Analytics of Lecture Materials available in SlideShare

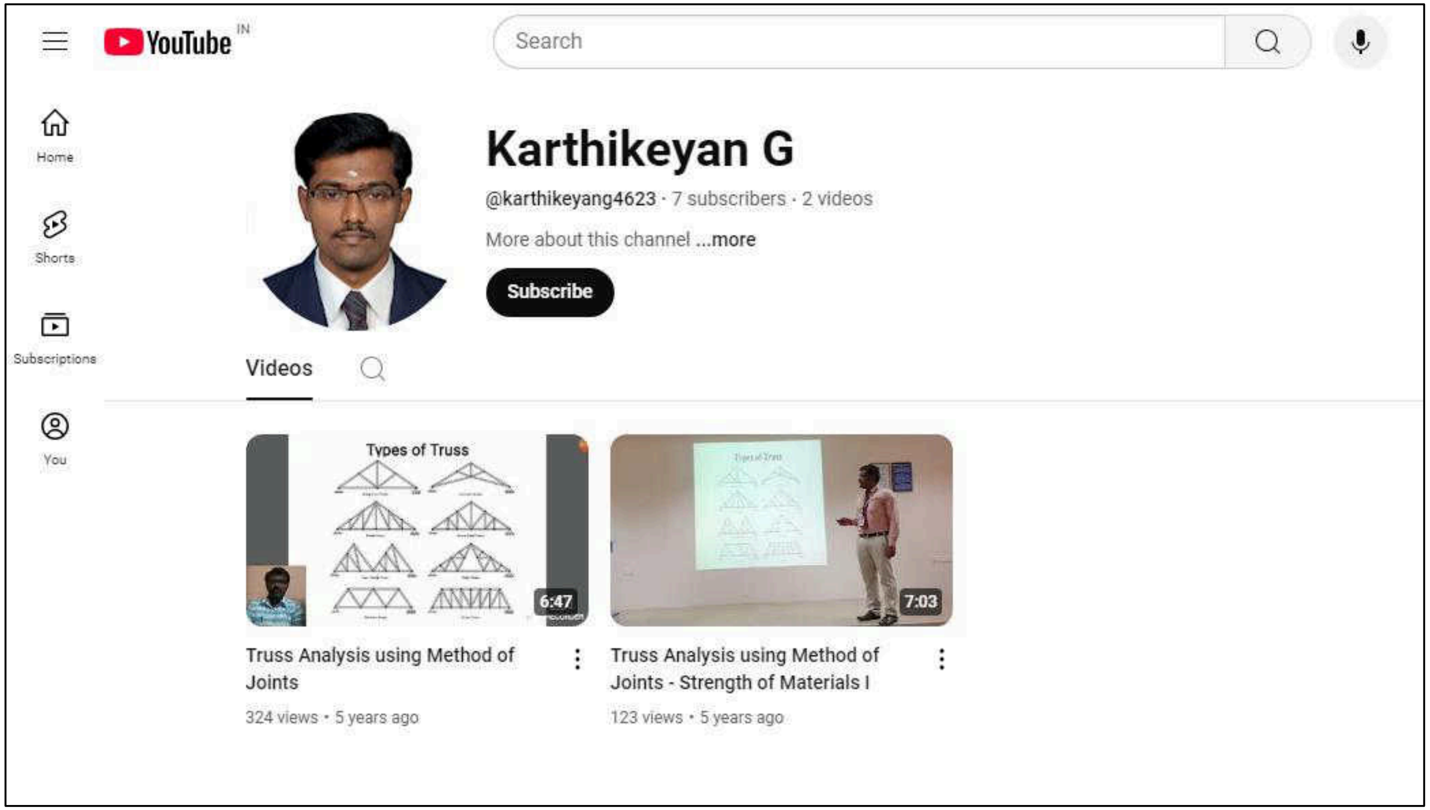
https://www.slideshare.net/chocka1



CE3503 Foundation Engineering (Google sites)

https://sites.google.com/ritrjpm.ac.in/foundationengineering/home

Fig. 5.5 Sample Analytics of Google Site – Foundation Engineering



Lecture Videos uploaded (YouTube)

https://www.youtube.com/@karthikeyang4623/videos

Fig. 5.6 Sample Analytics of Videos available in YouTube

The department and institute encourage faculty members to deliver guest lectures in National and International level programs conducted in RIT and other academic institutions. Delivering guest lectures helps faculty members to enrich their knowledge in their area of research and teaching by sharing and communicating with a larger audience. Feedbacks provided by the attendees helps to improve their teaching-learning process. The following **Table 5.4** provides the details of guest lectures delivered by the faculty of Civil Engineering.

Table 5.4 Guest Lectures delivered by Faculty Members

Name of Faculty	Type of Program	Title / Topic of lecture	Organized by	Date of lecture delivered
Mrs.Leema Margret	Webinar	Forces on Impounding Structure	Er.Perumal Manimekalai College of Engineering, Hosur	19.05.2021
Dr.M.S.Dharmar	Guest Lecture	Opportunities in Civil Engineering	Shanmuganathan Engineering College, Pudukkottai	03.09.2022
Mr.T.Chockalingam	Guest Lecture	Entrepreneurship and Innovation as Career Opportunity	RIT – IIC & RIT SIP Cell	09.11.2022
Mr.T.Chockalingam	Seminar	How to write a Proposal	PSR Engineering College, Sivakasi	06.07.2023
Mr.T.Chockalingam	Guest Lecture	Ideation: Cultivation Creativity for innovation	Ayya Nadar Janaki Ammal College, Sivakasi	25.03.2024
Dr.G.Karthikeyan	Guest Lecture	Career Opportunities in Civil Engineering	PACR Polytechnic College, Rajapalayam	14.08.2024

5.6. Faculty as participants in Faculty development/training activities/STTPs (15)

Name of the Faculty	Max. 5 per Faculty		
	2023-2024 (CAYm1)	2022-2023 (CAYm2)	2021-2022 (CAYm3)
Dr.S.Dharmar	5	5	3

Dr.M.Indhumathi	5	5	5
Dr.G.Karthikeyan	5	5	5
Mrs.C.Subha	5	5	5
Mrs.D.Darling Helen Lydia	0	5	5
Mr.T.Chockalingam	5	5	5
Mrs.R.Kalaimani	5	5	5
Mr.R.Muruganantham	5	5	5
Mr.J.Ram Prashath	0	0	3
Mr.A.Manicka Mamallan	5	5	5
Mrs.A.Leema Margret	5	5	5
Mr.V.Ragavan	5	5	5
Sum	50	55	56
RF= Number of Faculty required to comply with 20:1 Student-Faculty ratio as per 5.1	**9**	**9**	**9**
Assessment = 3 × (Sum/0.5RF) (Marks limited to 15)	**33.34**	**36.67**	**37.34**

Average assessment over 3 years:

5.7. Research and Development (30)

5.7.1. Academic Research (10)

The following table lists the core research areas of faculty in Civil engineering department.

Table 5.5 Core research areas

S.No.	Research Group	Faculty Members
1.	Structural Engineering	Dr.M.S.Dharmar Dr.M.Indhumathi Mr.T.Chockalingam Mrs.R.Kalaimani Mr.R.Muruganantham Mr.A.Manickamamallan Mrs.A.Leema Margret Mr.V.Ragavan
2.	Environmental Engineering	Mrs.C.Subha
3.	Infrastructure Engineering	Dr.G.Karthikeyan
4.	Hydrology and Water Resources Engineering	Mrs.B.Bharani Baanu
5.	Geotechnical Engineering	Mr.V.Jeevanantham

(A) Number of quality publications in refereed/SCI Journals, citations, Books/Book Chapters etc. (6)

1.Journal Publications:

The following table 5.6 list the paper publication in National/International journals by our faculty members

Table 5.6 List of Journal Publications

S. No.	Faculty Name	Title of the Paper	Name of Journal	Volume/ Issue	Year of Publication	SCI/ Scopus/ UGC	Impact Factor	Citation	ISSN/ E-ISSN No.
1.	Karthikeyan Ganesan, Vijai Kanagarajan, Jerlin Regin Joseph Dominic	Investigation on Corrosion and flexural behaviour of Reinforced Concrete using Marine Sand	Jurnal Teknologi (Sciences & Engineering)	87	2025	SCI	0.6	-	2180-3722
2.	Subha C Priya A.K.	Mustard (Brassica Nigra) Cake Powder Biochar/Nafion Composite as ORR Catalyst in Single Chambered Air Cathode MFC	Iranian Journal of Chemistry and Chemical Engineering	43/2	2024	SCI	1.0	-	1021-9986
3.	Chithambar Ganesh A Raju H.P Leema Margret A Jinendra U	Rice Husk Ash based Sodium Silicate as the Alkali Activator in slag based Geopolymer Concrete	E3S Web of Conferences	559	2024	Scopus	-	-	2267-1242
4.	Prakash R. Srividhya S Vijayalakshmi R. Muruganantham R	Characterization and performance evaluation of coconut shell concrete with alccofine supplements	Materials Protection/ Zastita Materijala	65/3	2024	Scopus	-	-	0351-9465

5.	Karthikeyan G Dharmar S Ragavan V	Study on Performance of Paver Block Using Prosopis Juliflora Ash	Global NEST	25/9	2023	SCI	1.4	1	1790-7632
6.	Karthikeyan G Dharmar S Ragavan V	Study on Performance of Paver Block Using Prosopis Juliflora Ash	Global NEST	25/9	2023	SCI	1.4	1	1790-7632
7.	Karthikeyan G Leema Margret A Muruganantham R	Influence of Utilizing Prosopis Juliflora Ash as Cement on Mechanical Properties of Cement Mortar And Concrete	Global NEST	26/1	2023	SCI	1.4	2	1790-7632
8.	Marimuthu Venkatesh Perumal Andavar Manicka Mamallan Veeriah Ragavan	Determination and eco-friendly suppression of fluoride contamination in ground water samples using activated carbon constituents of carica papaya – A natural adsorbent	Materials Today: Proceedings	In Press	2023	Scopus	-	-	2214-7853
9.	Vigneshkumar Chellappa Angelin Lincy Gunasekaran	Factors influencing construction waste generation: perspectives from India	Proceedings of the Institution of Civil Engineers -	177/3	2023	Scopus	0.7	2	1747-6526

	Kalaimani Ramakrishnan		Waste and Resource Management						
10.	Parthiban Loganathan Hemadri Prasad Raju A. Leema Margret V. Ragavan A. Chithambar Ganesh	Extraction of Heavy Metals from Soil Affected by Landfill Leachate through Constructed Wetlands: A Phytoremediation Approach to Rejuvenating the Contaminated Environment	E3S Web of Conferences	405	2023	Scopus	-	-	2267-1242
11.	P. Kunal Kaushik D. Justus Reymond C. Subha V. Lawrance	Review on Biocrude generated from plastic waste using Pyrolysis process	AIP Conference Proceedings	2852/1	2023	Scopus	-	-	1551-7616
12.	Leema Margret A, Karthikeyan G	Numerical study on flexural behavior of reinforced concrete beam using MATLAB	AIP Conference Proceedings	2831/1	2023	Scopus	-	2	1551-7616
13.	Subha Chandrasekarabarathi, Kalaimani Ramakrishnan, Justus Reymond David, Dharmar Sakkarai	Surface water analysis and purification system for Korampallam lake in Tuticorin	AIP Conference Proceedings	2831/1	2023	Scopus	-	-	1551-7616
14.	Sathish Kumar V, Dhivakar M, Nagamani S,	Removal of pharmaceuticals from wastewater: a	Global NEST	26/4	2024	SCI	1.4	1	1790-7632

	Dhanalakshmi A, Leema Margret A	review of different adsorptive approaches							
15.	B.G. Vishnuram, P. Muthupriya, A Dhanalakshmi, A. Leema Margret	Fly-ash and GGBS based geo-polymer concrete with granite powder as partial replacement of M-Sand for sustainability	Materials Today: Proceedings	In Press	2023	Scopus	-	5	2214-7853
16.	Karthikeyan G Dharmar S Ragavan V	Study on Performance of Paver Block Using Prosopis Juliflora Ash	Global NEST	25/9	2023	SCI	1.4	1	1790-7632
17.	T.Chockalingam C.Vijayaprabha J.Leon Raj	Experimental study on size of aggregates, size and shape of specimens on strength characteristics of pervious concrete	Construction and Building Materials	385	2023	SCI	7.693	20	0950-0618
18.	L Dhal, R Gopalakrishnan, S Dharmar, M Praveenaa	Hybrid Fibre Reinforced Concrete with Replacement of Fine Aggregate	Journal of the Balkan Tribological Association	28	2022	Scopus	-	3	1310-4772
19.	A.K. Priya C. Subha P. Senthil Kumar R. Suresh Saravanan Rajendran Yasser Vasseghian	Advancements on sustainable microbial fuel cells and their future prospects: A review	Environmental Research	210	2022	SCI	8.431	44	0013-9351

	Matias Soto-Moscoso								
20.	M.Indhumathi A.Leema Margret V.Ragavan J.Ram Prashath	Investigation on corrosion behaviour of geopolymer concrete using DMS and M−Sand as a fine aggregate under ambient curing conditions	Materials Today: Proceedings	In Press	2023	Scopus	-	1	2214-7853
21.	Dhanalakshmi Ayyanar B.G. Vishnuram P. Muthupriya M. Indhumathi Anbarasan	An experimental investigation on strength properties and flexural behaviour of ternary blended concrete	Materials Today: Proceedings		2023	Scopus		-	2214-7853
22.	S.Dharmar S.Nagan	Assessment of the characteristics of ferro-geopolymer composite box beams under flexure	Advances in Concrete Construction	15/4	2023	SCI	2.580	-	2287-5301
23.	C.Subha Priya ArunKumar Rajesh Banu Jeyakumar	Effect of organic loading on bioelectricity generation potential of dual-chambered microbial fuel cell treating chocolaterie wastewater	Environmental Progress & Sustainable Energy	42/4	2023	SCI	2.824	1	1944-7450
24.	M.Indhumathi V.Ragavan Darling Helen Lydia D	A Study on Non Destructive Tests And Flexural Behaviour Of Geopolymer	E3S Web of Conferences	387	2023	Scopus	-		2267-1242

		Concrete Using Different Fine Aggregates							
25.	Karthikeyan G Leema Margret A	Experimental study on mechanical properties of Textile Reinforced Concrete (TRC)	E3S Web of Conferences	387	2023	Scopus	-	3	2267-1242
26.	R. Kalaimani C. Subha D. Justus Reymond C. Vignesh kumar	Investigation on Strength Characteristics of Self Compacting Concrete incorporated with AR Glass Fibers	E3S Web of Conferences	387	2023	Scopus	-	2	2267-1242
27.	Kalaimani Ramakrishnan Vigneshkumar Chellappa Subha Chandrasekarabarathi	Manufacturing of Low-Cost Bricks Using Waste Materials	Materials Proceedings	13/1	2023	Scopus	-	10	2673-4605
28.	A. Dhanalakshmi J. Jeyaseela S. Karthika A. Leema Margret	An Experimental Study on Concrete with Partial Replacement of Cement by Rice Husk Ash and Bagasse Ash	E3S Web of Conferences	387	2023	Scopus	-	9	2267-1242
29.	R Gopalakrishnan, S Dharmar, D Purushothaman	Production of Construction Material Using Construction and Demolition Waste by Geo-	Journal of the Balkan Tribological Association	27/6	2021	Scopus		-	1310-4772

		Polymerisation of Industrial Waste							
30.	S.Dharmar S.Nagan	Strength Behavior of Flat and Folded Fly Ash-Based Geopolymer Ferrocement Panels under Flexure and Impact	Advances in Civil Engineering	2021	2021	SCI	1.843	-	1687-8094
31.	Karthikeyan Ganesan Vijai Kanagarajan Jerlin Regin Joseph Dominic	Influence of marine sand as fine aggregate on mechanical and durability properties of cement mortar and concrete	Materials Research Express	9/3	2022	SCI	2.025	17	2053-1591
32.	G. Murali Sallal R. Abid K. Karthikeyan M.K.Haridharan Mugahed Amran A. Siva	Low-velocity impact response of novel prepacked expanded clay aggregate fibrous concrete produced with carbon nano tube, glass fiber mesh and steel fiber	Construction and Building Materials	284	2021	SCI	4.419	67	0950-0618
33.	S.Dharmar R. Gopalakrishnan A. Mohan	Environmental Effect of Denitrification of Structural Glass by coating Tio_2	Materials Today: Proceedings	45/7	2021	Scopus	-	47	2214-7853

34.	M.K.Haridharan	Low-velocity impact response of novel prepacked expanded clay aggregate fibrous concrete produced with carbon nano tube, glass fiber mesh and steel fiber	Construction and Building Materials	284	2021	SCI	4.419	49	0950-0618
35.	S.Dharmar	Environmental Effect of Denitrification of Structural Glass by coating Tio_2	Materials Today: Proceedings	45/7	2021	Scopus	-	43	2214-7853
36.	T. Chockalingam	Strength and abrasion characteristics of pervious concrete	Road Materials and Pavement Design	21/8	2019	SCI	2.582	24	1468-0629
37.	G.Karthikeyan	Structural Rehabilitation and Strengthening of Column using Micro Concrete and Additional Reinforcement	International Journal of Engineering Research & Technology	7/6	2019	Non-Scopus	-	01	2278-0181
38.	G.Karthikeyan	Experimental Study on Concrete by Replacement of Fine Aggregate with Copper Slag, GGBS and M-Sand	International Journal of Innovative Science and Research Technology	3/3	2018	-	-	02	2456-2165

39.	G.Karthikeyan	Doubling of Tracks in Plate Girder Bridges without Demolition	Journal of Civil and Construction Engineering	4/1	2018	-	-	-	2457-001X
40.	Manicka Mamallan A	Experimental Study on Reinforced Concrete Beam Strengthened & Retrofitted with GFRP Sandwich Panels	International Journal of Applied Engineering Research	13/9	2018	-	-	-	0973-4562
41.	Muruganantham R	Experimental Study on Concrete Thermal Conductivity by incorporating copper and steel Slag	Journal of Ceramics and Concrete Sciences	3/1	2018	-	-	-	2582-1938
42.	Ram Prashath J								
43.	Kalaimani R								
44.	A.Hari Rama Lakshmi	Effect of Jute on Properties of Lime and Baggase Ash Stabilized Black Soil	International Research Journal of Engineering and Technology	5/4	2018	-	-	-	2395-0072
45.	C.Subha	Prediction of Landslides in Nilgris	Journal of Geotechnical Studies	3/1	2018	-	-	-	2581-9763

Citations:

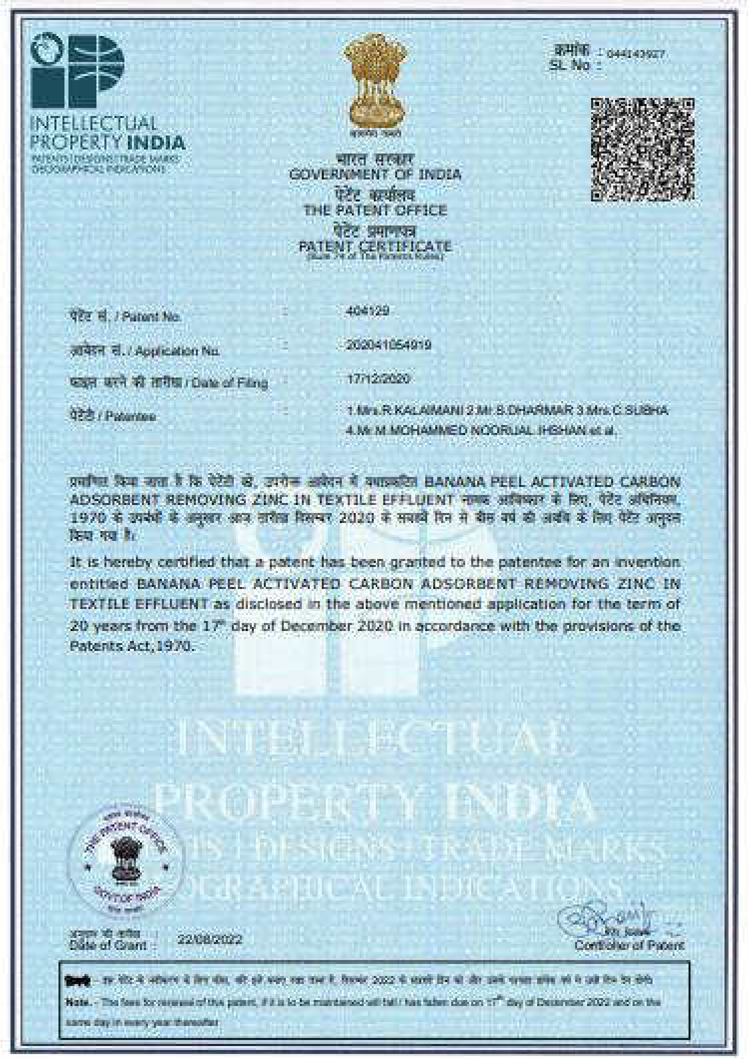
Table 5.7 Citation Details

S. No.	Name of the Faculty Member	Google Scholar			Scopus Index	
		H Index	i-10 Index	Total Citations	H Index	Total Citations
1.	Dr.S.Dharmar	7	4	168	3	39
2.	Dr.M.Indhumathi	2	1	32	2	25
3.	Dr.G.Karthikeyan	3	1	32	2	18
4.	Mrs.C.Subha	4	3	146	3	118
5.	Mr.T.Chockalingam	2	2	55	2	47
6.	Mrs.R.Kalaimani	2	1	14	1	2
7.	Mr.R.Muruganantham	2	0	4	1	1
8.	Mrs.A.Leema Margret	3	0	23	2	9
9.	Mr.V.Ragavan	1	0	5	1	2
10.	Mrs.B.Bharani Banu	4	1	39	1	7
11.	Mr.V.Jeevanantham	3	0	24	3	13

2.Patent granted & IPR Registered

Table 5.8 Patent Details

S. No.	Title of the Invention	Application Number	Date of Filing	Date of Publication	Current Status
1.	Banana Peel Activated Carbon Adsorbent Removing Zinc in Textile Effluent	202041054919	17.12.2020	25.12.2020	Granted on 22.08.2022



INTELLECTUAL PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS
GEOGRAPHICAL INDICATIONS

क्रमांक : 044143927
SL No :

भारत सरकार
GOVERNMENT OF INDIA
पेटेंट कार्यालय
THE PATENT OFFICE
पेटेंट प्रमाणपत्र
PATENT CERTIFICATE
(Rule 74 of The Patents Rules)

पेटेंट सं. / Patent No.	:	404129
आवेदन सं. / Application No.	:	202041054919
फाइल करने की तारीख / Date of Filing	:	17/12/2020
पेटेंटी / Patentee	:	1.Mrs.R.KALAIMANI 2.Mr.S.DHARMAR 3.Mrs.C.SUBHA 4.Mr.M.MOHAMMED NOORUAL IHSHAN et al.

प्रमाणित किया जाता है कि पेटेंटी को, उपरोक्त आवेदन में यथाप्रकटित BANANA PEEL ACTIVATED CARBON ADSORBENT REMOVING ZINC IN TEXTILE EFFLUENT नामक आविष्कार के लिए, पेटेंट अधिनियम, 1970 के उपबंधों के अनुसार आज तारीख दिसम्बर 2020 के सत्रहवें दिन से बीस वर्ष की अवधि के लिए पेटेंट अनुदत्त किया गया है।

It is hereby certified that a patent has been granted to the patentee for an invention entitled BANANA PEEL ACTIVATED CARBON ADSORBENT REMOVING ZINC IN TEXTILE EFFLUENT as disclosed in the above mentioned application for the term of 20 years from the 17th day of December 2020 in accordance with the provisions of the Patents Act,1970.

अनुदान की तारीख :
Date of Grant : 22/08/2022

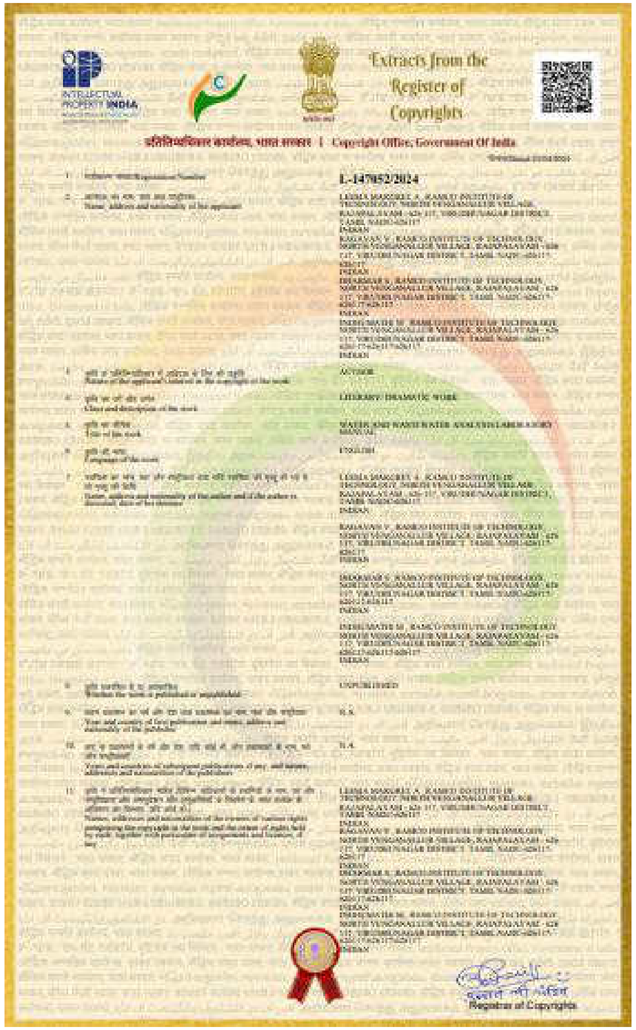
Controller of Patent

Note. - The fees for renewal of this patent, if it is to be maintained will fall / has fallen due on 17th day of December 2022 and on the same day in every year thereafter.

Figure 5.7 Patent Certificate

Table 5.9 Copyright status of the Laboratory Manuals

S.No.	Title of Laboratory Manual	Authors	Diary Number	Status
1.	CE3311 - Water and Wastewater Analysis Laboratory	Mrs.A.Leema Margret , AP/ Civil Mr.V.Ragavan, AP/Civil Dr.S.Dharmar, ASP(HoD)/Civil Dr.M.Indhumathi, ASP/Civil	34421/2023-CO/L	Registered
3.	CE3411 - Hydraulic Engineering Laboratory	Mrs.B.Bharani Baanu, AP/Civil Dr.S.Dharmar, ASP(HoD)/Civil Mrs.C.Subha, AP(SG) / Civil Mrs.R.Kalaimani, AP/Civil	24101/2024-CO/L	Applied
4.	CE3412 - Materials Testing Laboratory	Mr.V.Ragavan, AP/Civil Dr.S.Dharmar, ASP(HoD)/Civil Mrs.A.Leema Margret , AP/ Civil	34427/2024-CO/L	Applied
5.	CE3413 - Soil Mechanics Laboratory	Mr.V.Jeevanantham, AP/Civil Dr.S.Dharmar, ASP(HoD)/Civil Dr.M.Indhumathi, ASP/Civil	24502/2024-CO/L	Applied
6.	CE3511- Highway Engineering Laboratory	Mr.T. Chockalingam, AP(SG)/Civil Dr.S.Dharmar, ASP(HoD)/Civil Mr.V.Ragavan, AP/Civil	35205/2024-CO/L	Applied



Extracts from the Register of Copyrights

Copyright Office, Government Of India

L-147852/2024

LITERARY/ DRAMATIC WORK

ENGLISH

UNPUBLISHED

N.A.

N.A.

Registrar of Copyrights

Fig. 5.8 Registered Copyright for Water and Wastewater Analysis Laboratory

3.Books/Book Chapters

Table 5.10 Details of Books/Book Chapters

S. No.	Faculty Name	Title of Book	Name of Publisher	DOI/ISBN	Year of Publication	Language	Book/Book Chapter
1.	Dr.M.Indhumathi	Estimation, Costing and Valuation Engineering	Magnus Publications	978-81-972240-5-8	2024	English	Book
2.	Dr.M.Indhumathi	Construction Materials and Technology	Sruthi Publishers	13:978-81-958745-0-7	2022	English	Book
3.	Dr.M.Indhumathi	Geopolymer Concrete under Ambient Curing	Intech Open	10.5772/intechopen.97541	2021	English	Book Chapter

4.Conference Publications:

Table 5.11 List of Conference Publications

S. No.	Faculty Name	Title of the Paper	Name of Conference	Year of Publication	Date of Conference	ISBN
1	Dr.M.Indhumathi Mr.V.Ragavan	Experimental Investigation on Fibre Reinforced Geopolymer Concrete Geopolymer Concrete	2023 Third Global Conference on Recent Advances in Sustainable Materials (GC-RASM 2023)	2023	27th and 28th July 2023	-
2	Mrs.R.Kalaimani	Experimental Study on Performance of Cement-Based Batteries	2nd International Conference on WATER, ENERGY & ENVIRONMENT [WEECON 2023]	2023	29th and 30th December 2023	-
3	Dr.M.Indhumathi	Experimental Investigation on Concrete Using Industrial By-Products	Second International Conference on Recent Trends in Engineering and Technology (ICRTET 2024)	2024	14th and 15th March 2024	-
4	Mr.T.Chockalingam	Experimental Investigation on Strength and Hydrological Behaviour of Geo-Gird Reinforced Pervious Concrete	International Conference on Recent Trends in Engineering & Science (ICRTES-2024)	2024	2nd and 3rd May 2024	-
5	Dr.S.Dharmar Dr.G.Karthikeyan Mr.V.Ragavan	Experimental Study on Paver Block Using Prosopis Julifora Ash	International Conference on Sustainable Technology in Civil Engineering and Applied Sciences-2023	2023	24th and 25th March 2023	978-93-5759-825-5
6	Mr.A.Manicka Mamallan	Analysis Of Natural Coolant on The Rooftop To Lower Room Temperature	International Conference on Sustainable Technology in Civil Engineering and Applied Sciences-2023	2023	24th and 25th March 2023	978-93-5759-825-5
7	**Mr.V. Ragavan** **Mr.A. Manicka Mamallan**	Study and Evaluation of Firecracker-related Soil Contamination in Selected areas around Sivakasi	International Conference on Sustainable Technology in Civil Engineering and Applied Sciences-2023	2023	24th and 25th March 2023	978-93-5759-825-5
8	Chockalingam T	A Review on Permeability of Pervious Concrete	International Conference on Smart Technologies and Applications-2022	2022	11th and 12th March 2022	978-93-5593-352-2

9	A.Leema Margret V.Ragavan	Investigation of Mechanical Properties of High-Performance Concrete using Industrial Waste with Well Graded Aggregates	International Conference on Materials Science and Manufacturing Technology	2022	8^{th} - 9th April 2022	-
10	D.Darling Helen Lydia	Performance Study on Damaged Cylinder with SFRP Under Various Loading Conditions using Abaqus	International Conference on Smart Technologies and Applications-2022	2022	11^{th} and 12^{th} March 2022	978-93-5593-352-2
11	C.Subha	Investigation on Industrial Effluent Treatments Using Dual Chamber Microbial fuel Cells	Proceedings of Second International Conference on Advances in Management and Technology	2021	11^{th} -12^{th} December 2021	978-93-5426-863-2
12	C.Subha R.Kalaimani	Microbial Fuel Cells for Treating Seafood Processing Wastewater	International Conference on Recent Trends in Applied Sciences and Computing Engineering	2021	17^{th} – 19^{th} December 2021	-
13	Chockalingam T	Real Time Monitoring of Cracks using Deep Learning Techniques	National Web Conference on Challenges and Innovation in Engineering and Technology	2021	19^{th} and 20^{th} March 2021	-
14	Chockalingam T	Prediction of Properties of Pervious Concrete using Multiple Linear Regression	1^{st} International Conference on Advances in Construction Materials and Management	2021	25^{th} and 26^{th} March 2021	-
15	Karthikeyan G	Effect of Geometry of Core on the behaviour of Hollow Core Beam	National Web Conference on Challenges and Innovation in Engineering & Technology	2021	19^{th} and 20^{th} March 2021	-
16	Karthikeyan G	Effect of Geometry on the behaviour of RCC Column	National Web Conference on Challenges and Innovation in Engineering & Technology	2021	19^{th} and 20^{th} March 2021	-
17	R.Kalaimani C.Subha	Experimental Study on the Effect of Magnetized Water on Compressive Strength of Concrete	International Web Conference on Smart Engineering Technologies	2020	26^{th} &27^{th} June 2020	-

18	R.Kalaimani C.Subha	Experimental Study on Utilization of Treated and Magnetized Industrial Wastewater in Strength of Concrete	Second International Conference on Advanced Materials Chemistry at the interfaces of Energy, Environment and Medicine – AMCI – 2020	2020		-
19	C Subha	Textile Effluent Treatment and Electricity Generation Using Microbial Fuel Cell	International Conference on Advances in Management and Technology	2020	6^{th} – 7^{th} November 2020	-
20	C Subha	Anaerobic Microbial Fuel Cell for Treating Complex Organic Wastewater	International Conference on Advances in Management and Technology	2020		-
21	C Subha R Kalaimani	Experimental Study on Water Softening Using Biochar	National Web Conference on Challenges and Innovation in Engineering &Technology	2021	19^{th} and 20^{th} March 2021	-
22	Leema Margret A	Analytical Investigation on Flexural Behaviour of RC beam strengthened with CFRP using ABAQUS	National Web Conference on Challenges and Innovation in Engineering &Technology	2021	19^{th} and 20^{th} March 2021	-
23	Ragavan V	Numerical Simulation on Precast T-Beam Using ETABS	National Web Conference on Challenges and Innovation in Engineering &Technology	2021	19^{th} and 20^{th} March 2021	-
24	Dharmar S	Numerical Simulation on Flexural Behaviour of Reinforced Concrete Beam by Varying Cover Depth Under Monotonic Loading	National Web Conference on Challenges and Innovation in Engineering &Technology	2021	19^{th} and 20^{th} March 2021	-
25	Darling Helen Lydia D	Treatment of Industrial Wastewater using Green Synthesis of Copper Nano Particles	National Web Conference on Challenges and Innovation in Engineering &Technology	2021	19^{th} and 20^{th} March 2021	-
26	Ram Prashath J	Analysis of Impact Behaviour of Masonry Wall Using Abaqus	National Web Conference on Challenges and Innovation in Engineering &Technology	2021	19^{th} and 20^{th} March 2021	-

27	R.Muruganantham	Behaviour of Masonry Wall Under Blast Loading	National Web Conference on Challenges and Innovation in Engineering &Technology	2021	19th and 20th March 2021	-

(B) PhD guided /PhD awarded during the assessment period while working in the institute

1.Research Centre Details

Table 5.12 Research Centre Details

S. No.	Name of the Department	Recognized R&D Centre Number	University
1	Civil Engineering	4467805	Anna University, Chennai

2. Ph.D. Guidance

Table 5.13 Research Supervisor details

S. No.	Name of the Supervisor	Supervisor Number	University
1.	Dr.S.Dharmar	4210005	Anna University, Chennai
2.	Dr.M.Indhumathi	4010010	Anna University, Chennai
3.	Dr.G.Karthikeyan	4310015	Anna University, Chennai

3.Research Scholar Details:

Table 5.7.10 Research scholar details

S. No.	Name of the Faculty	Name of the Scholar	Registration Number	Mode	Area of Research	Year of registration	Status

1.	Dr.S.Dharmar	Mr.Venkateswaran	24121897107	Part Time	Structural Engg.	2024	Doing Coursework
2.	Dr.M.Indhumathi	Ms.Rebekhal D	24241897128	Part Time	Structural Engg.	2024	Doing Coursework
3.	Dr.M.Indhumathi	Mrs.Manju A	24236797104	Part Time	Arch.	2024	Doing Coursework
4.	Dr.M.Indhumathi	Mr.Muthuramu B	24131891200	Part Time	Structural Engg.	2024	Doing Coursework
5.	Dr.M.Indhumathi	Mr.Ramesh V	24141891199	Part Time	Structural Engg.	2024	Doing Coursework

4.Faculty members with Ph.D.

Table 5.14 Faculty members with Ph.D.

S. No.	Name of the Faculty	University	Ph.D. Awarded (Year)
1	Dr.S.Dharmar	Anna University	2022
2	Dr.M.Indhumathi	Anna University	2021
3	G.Karthikeyan	Anna University	2023

5. Faculty members pursuing Ph.D.

Table 5.15 Faculty members pursuing Ph.D.

S. No.	Name of the Faculty	University	Status
1.	Mrs.C.Subha	Anna University	Thesis submitted
2.	Mr.T.Chockalingam	Anna University	Thesis submitted

3.	Mrs.R.Kalaimani	Anna University	Confirmation completed
4.	Mr.R.Muruganantham	Anna University	Coursework Completed
5.	Mr.A.Manicka Mamallan	Anna University	Coursework Completed
6.	Mrs.A.Leema Margret	Anna University	Confirmation Completed
7.	Mr.V.Ragavan	Anna University	Confirmation completed
8.	Mrs.B.Bharani Baanu	Anna University	Thesis submitted

5.7.2. Sponsored Research (5)

2023-2024 (CAYm1)

Project Title	Duration	Funding Agency	Amount (Rs.)
Figure Diminishing Geopolymer Products	1 Year	Ministry of MSME	1400000.00
Sustainable Light-weight Building Block (SLBB) and its Manufacturing Method	1 Year	Ministry of MSME	900000.00
Experimental study on the future of sustainable building using textile reinforced concrete (TRC)	6 Months	Tamil Nadu Council for Science and Technology	7500.00
		Total	2307500.00

2022-2023 (CAYm2)

Project Title	Duration	Funding Agency	Amount (Rs.)
Eco friendly light weight bricks	6 Months	Tamil Nadu Council for Science and Technology	7500.00

2021-2022 (CAYm3)

Project Title	Duration	Funding Agency	Amount (Rs.)
Cement Battery	6 Months	Tamil Nadu Council for Science and Technology	7500.00

Cumulative Amount (X + Y + Z) = Rs.2322500.00

5.7.3. Development activities (10)

(A) Product Development:

Table 5.16 Details of Product Developed

S. No.	Name of the Faculty and Students	Type of the Product	Applications	Academic Year
1.	Mrs.C.Subha Mrs.R.Kalaimani Balaji Devar Piran M Arun Kumar S	Household Waste Composter	It will convert kitchen waste into manure	2023-2024
2.	Dr.G.Karthikeyan Suresh Anand J Bhasheer Mohamad M	Textile Reinforced Concrete	It can be used for Structural Elements, Facades and Thin-shell Structures, Retrofitting and Strengthening, Precast Elements Noise Barriers, Sustainability Initiatives, Pavements and Roadways, Footbridges and Pedestrian Walkways, Molds and Formworks, Artistic and Architectural Designs	2023-2024

S. No.	Name of the Faculty and Students	Type of the Product	Applications	Academic Year
3.	Mr.T.Chockalingam P.Aakash	Pervious Concrete	It can be used for Parking Lots, Driveways, Sidewalks and Pathways, Roadways, Stormwater Management, Green Building Projects, Paving for Parks and Recreational Areas, Pool Decks Low-Impact Development (LID) Solutions, Landscaping Features	2022-2023
4.	Mr.G. Karthikeyan M.Paartha Sarathi	Paver Blocks using supplementary cementitious material (Prosopis Juliflora)	Promising sustainable solution for the construction industry in the production of paver block by replacing cement	2022-2023
5.	Mrs.C.Subha Abdhur Rahman, K.Dinesh	Microbial Fuel Cell	A Single Chamber Microbial Fuel Cell is employed for treating distillery wastewater with simultaneous power generation.	2021-2022
6.	Dr.S.Dharmar Mrs.R.Kalaimani G.Mohanrajan S.Srivarshan M.RA.Suresh Kumar	Cement Battery	Cement matrix batteries are cost-effective because they generate electricity by interacting with water, which contributes to the development of sustainable energy solutions.	2021-2022
7.	Mrs.A.Leema Margret Balaji B Charu Mahesh M Mahesh Murugan R Parthiban M	Translucent Concrete	It observes the natural light source during the daytime and discharge the same during the night time, which leads to energy conservation.	2020-2021

(B) Research laboratories:

A workplace to conduct research to create products to solve problems related to sustainability. By means of that faculty members and students will nourish their Research and development activities specifically providing opportunities for the students to participate in Cube contest, design competition, project contest. Also, it provides opportunities to publish journals and conferences.

In Ramco Institute of Technology, Rajapalayam Department of Civil Engineering is having four research laboratories they are as follows:

- ❖ Project & Consultancy Lab
- ❖ RIT-Centre of Excellence for Building Information Modelling (BIM)
- ❖ RIT ICT Academy Centre of Excellence for Design Powered by Autodesk
- ❖ Centre for Geospatial Technology

1.Project & Consultancy Lab

Project and consultancy laboratory was established in the year 2022.This laboratory is mainly focused on enhancing the project work of the students as well as Faculty those who are doing the research works. The major equipment in the laboratory includes Rapid Chloride Penetration Test equipment, Concrete Impact testing equipment, Rebound Hammer, Total Station, Line Laser Level, Standard Penetration test equipment, Rapid Moisture Meter. In addition, various consultancy works like Field Density Test, Bearing Capacity of Soil, Topographical Survey, Compressive strength of existing structures etc., have been carried with the help of our faculty members. Project lab facilities are shown in Figure 5.9

Figure 5.9 Project Lab Facilities

Table 5.17 List of Projects

S.No.	Name of the Project	Type & Relevance of Project
(CAYm1 2023-2024)		
1	Mechanical Behavior of Roller compacted concrete using copper slag	Structural Engineering (Product)
2	Household Waste Composter	Environmental Engineering (Product)
3	Experimental investigation of High strength concrete using mill scale tailings	Structural Engineering (Product)
4	Experimental investigation of Geological Light weight Geopolymer Concrete Blocks	Structural Engineering (Product)
5	Influence of heat on mechanical properties of pervious concrete with different aggregate sizes	Structural Engineering (Product)

6	Experimental Investigation on mechanical properties of Geomortar-Ready Mix Powder	Structural Engineering (Product)
7	Experimental study on flexural behavior of textile reinforced concrete	Structural Engineering (Product)
8	Experimental study on lantana camera reinforced concrete	Structural Engineering (Product)
9	Experimental Investigation on carbon fiber reinforced geopolymer concrete	Structural Engineering (Product)
(CAYm2 2022-2023)		
1	Flexural Performance of Geogrid reinforced pervious Concrete beam with different aggregate sizes.	Structural Engineering (Product)
2	Utilization of LECA in GGBS based light weight geopolymer concrete with addition of lime	Structural Engineering (Product)
3	Self-Compacting Concrete using mill scale waste	Structural Engineering (Product)
4	A comparative study on multiblock	Structural Engineering (Product)
5	Impact and Mechanical Properties of carbon fibre reinforced Concrete.	Structural Engineering (Product)
6	Eco-friendly light weight bricks	Structural Engineering (Product)
7	Experimental Investigation on mechanical behavior of fiber-based Roller compacted Concrete and copper slag	Structural Engineering (Product)
8	Study on LEED credits and Certification	Infrastructure Engineering (Product)
9	Experimental Study on Paver Block using Prosopis Juliflora Ash	Structural Engineering (Product)
10	Effect of Specimen size on Compressive strength of pervious concrete with different aggregate sizes	Structural Engineering (Product)
11	Influence of wood ash and fly ash based light weight geopolymer concrete using LECA	Structural Engineering (Product)
12	Impact behavior of GPC Using DMS and M-Sand as fine aggregates	Structural Engineering

		(Product)
13	Experimental Study on mechanical properties of Textile Reinforced Concrete	Structural Engineering (Product)
14	Treatment of textile effluent in palm fruit cell	Environmental Engineering (Product)
15	Experimental study on the strength behavior of lantana camara reinforced concrete beam	Structural Engineering (Product)
16	Experimental Investigation of Arsenic Removal from water using Iron Oxide Nanoparticles Incorporated in Activated Carbon Synthesized from Natural adsorbents (Grass Cuttings)	Environmental Engineering (Product)
17	Experimental Investigation on stabilization of Black Cotton soil using Borosilicate Glass Powder	Geotechnical Engineering (Product)
(CAYm3 2021-2022)		
1	Experimental study on Microbial fuel cell performance on pollutant removal by utilizing sesame oil cake as Bio-char catalyst	Environmental Engineering (Product)
2	Experimental investigation on packing density of different fine aggregate	Structural Engineering (Product)
3	Experimental Investigation on Mechanical behavior of fibre-based Roller Compacted Concrete	Structural Engineering (Product)
4	Experimental study on Gypsum panel and blocks incorporated with waste recycled plastic	Structural Engineering (Product)
5	Experimental study on mechanical property of mill scale based self-compacting concrete	Structural Engineering (Product)
6	Experimental Study of Performance of Cement-Based Batteries	Structural Engineering (Product)
7	Effect of PVA Fiber on Slag Concrete	Structural Engineering (Product)
8	Experimental Investigation on Stabilized Mud Block incorporated with Copper Slag	Structural Engineering (Product)
9	Experimental investigation on flexural behavior of marine sand based Reinforced Cement Concrete Beam	Structural Engineering (Product)
10	Experimental study on Compressive strength and durability properties of pervious concrete	Transportation Engineering (Product)

2.RIT-Centre of Excellence for Building Information Modelling (BIM)

The Department of Civil Engineering, Ramco Institute of Technology, Rajapalayam has established the **RIT Centre of Excellence for Building Information Modeling (BIM - A Futuristic Lab) in collaboration with Bentley Education and TechApps Consulting, Chennai** with cutting-edge industry-relevant software catering to diverse sub-domains of Civil Engineering. A MoU was signed between **RIT and TechApps for Bentleys software training and knowledge transfer among students and faculty members**. RIT BIM Laboratory has equipped with the Bentley Softwares like STAAD.Pro Connect Edition, Open Buildings Designer, Open Roads Designer and other products available for academics from Bentley Education which facilitates the students to do simulation and projects. Through the Centre of Excellence, Guest lectures, and Value Added Courses conducted periodically for the benefit of students and faculty members.

Fig. 5.10 MoU Between RIT and TechApps Consulting & BIM Lab

Table 5.18 Activities organized through RIT-CoE for BIM

S.No	Title of the Event	Date & Duration	Resource Person

1.	Virtual Webinar on Reality Modeling	18.05.2022 & 4 hours	Mr.S. Rajesh Kumar, Director, Technical Services, TechApps Consulting, Chennai.
2.	Standard Operating Procedure for Bentley Certification	19.05.2022 & 4 hours	Mr.S. Rajesh Kumar, Director, Technical Services, TechApps Consulting, Chennai.
3.	Hands on Training on Building Information Modelling	23.07.2024-27.07.2024	Mr.Mohammed Muneeb, Senior BIM Consultant Mr.S. Rajesh Kumar, Director Technical Services, TechApps Consulting, Chennai.
4.	Big Benefits of Building Information Modelling (BIM)	22.04.2022 & 2 hours	Chandra Shekar, Co-Founder of Turn BIM Engineering Services
5.	Hands on Training on Project Planning and Management using Primavera and MS Project	25.10.2024 to 26.10.2024 2 Days	Er.K.Iyappan, Project Lead, Tech Mahindra India Pvt. Ltd., Bengaluru
6.	Hands on Training on STAAD.Pro Connect Edition V22	27.06.2023 and 28.06.2023 2 Days	Mr.R.Ganesh Kumar, Sr.Structural Consultant, TechAppss, Consulting, Chennai
7.	Value Added Course on Open Building Designer	25.08.20222 & 26.08.2022 2 Days	Mr.Akbar Ali Khader & Mr.S.Rajesh Kumar, TechAppss, Chennai
8.	Hands on Training on “STAAD.Pro Connect Edition V22”	27.01.2023 to 28.01.2023 2 Days	Mr.R.Ganesh Kumar, Sr.Structural Consultant, TechAppss, Consulting, Chennai

2.RIT ICT Academy Centre of Excellence for Design Powered by Autodesk

The Department of Civil Engineering, Ramco Institute of Technology, Rajapalayam, has established the **RIT Centre of Excellence for design powered by Autodesk in collaboration with ICT Academy** for cutting-edge industry-relevant software catering to diverse sub-domains of Civil Engineering. RIT Design laboratory has equipped with the Autodesk software like Autodesk Architecture, Engineering & Construction

Collection, Robot Structural Analysis professionals, Advance Steel, Structural bridge Design, Navis work, Fabrication CADmep, AutoCAD Takeoff, Geospatial Infraworks. The Autodesk provides students with a set of BIM and CAD tools supported by a cloud-based common data environment that facilitates project delivery from early-stage design through to construction. The Department of Civil Engineering, Ramco Institute of Technology, Rajapalayam, has established the RIT Centre of Excellence for design powered by Autodesk in collaboration with ICT Academy during the Academic year 2023-2024 for cutting-edge industry-relevant software catering to diverse sub-domains of Civil Engineering.

Fig.5.11 ICT Academy - Centre of Excellence for Design – Certificate

Table 5.19 Activities organized through RIT ICT Academy - COE

S.No	Title of the Event	Date & Duration	Resource Person
1.	Revit Architecture – Building Information Modelling	07.10.2024 to 11.07.2024 5 Days	Ms.Priya Dharshika Technical Trainer Training and Development ICT Academy

3.Centre for Geospatial Technology

Centre for Geospatial Technology was established in the year 2024 in association with Land Coordinates Technology (LCT), Chennai to provide hands-on experience to students and faculty members on Differential Global Positioning System (DGPS) and Drone Surveying. This hands-on experience helps them to enhance their skills regarding the cutting-edge technologies happening in the Geospatial sector. Through this center, our department completed the following activities which were helpful for the students and society.

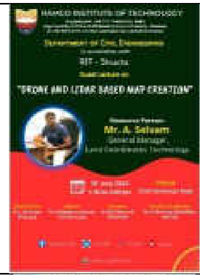
Fig. 5.12 MoU Between RIT and Land Coordinates Technology

Our department faculty members and students engaged in preparing the topographic layout of Kondaneri lake which is situated in city of Rajapalayam in association with Land Coordinates Technology (LCT), Chennai.

Fig. 5.13 Topographical Survey – Kondaneri Lake

Table 5.20 Activities organized

S. No	Title of the Event	Date & Duration	Remarks
1.	Drone and LIDAR based Map Creation	28-07-2023, 1.30 pm – 3.00 pm	
2.	Workshop on Total Station and DGPS	11-10-2023 To 12-10-2023 09.00 am – 04.00 pm	

2.	One Day Hands On Training On Mapping Using DGPS	24-07-2024, 10.00 am – 04.00 pm	
3.	One Day Hands on Training on Drone Surveying	24-10-2024 10.00 am – 04.00 pm	

(C) Instructional materials:

Instructional materials refer to the resources and tools that are used by the faculty to facilitate learning. This includes textbooks, laboratory equipment, software, multimedia resources, Models and Prototypes and other teaching aids.

- Faculty of civil engineering provide students with extensive hand-out for all the courses consisting of important contents of the subject, worked examples, etc., which helps the students to revise the contents during the exams and also understand the subjects in a simpler manner.
- Laboratory Manuals and necessary instructional materials for all the laboratory courses are developed by the respective faculty members for use by the students in order for the better understanding of the laboratory experiments and also to enrich their practical skills and knowledge.

Table 5.21 List of Laboratory Manuals

S.No	Title of the Laboratory Manual
1.	CE3311 - Water and Wastewater Analysis Laboratory
2.	CE3411 - Hydraulic Engineering Laboratory
3.	CE3412 - Materials Testing Laboratory
4.	CE3413 - Soil Mechanics Laboratory
5.	CE3511- Highway Engineering Laboratory
6.	CE3361 - Surveying and Levelling Laboratory
7.	CE3611 - Building Drawing and Detailing Laboratory
8.	CE3011 - Digitalized Construction Lab

Library

The institution has a Main Library which is a very good resource center for teaching, learning & research. The main library has a widespread space with good seating capacity, state of art digital content. Reference Section, Circulation Counter, OPAC Search, Journals/Magazines and Newspaper Section are available in the main library. The main library holds a hybrid collection of printed/electronic resources which include books, journals, CDs/DVDs, e-books, previous years question papers, and student project reports. E-learning resources are available to facilitate the access of journal papers and enhance the knowledge of students and faculty. Some of the digital content include:

Science Direct	https://www.sciencedirect.com/
IEEE	https://ieeexplore.ieee.org/Xplore/home.jsp
DELNET	https://discovery1.delnet.in/

IEI	https://www.ieindia.org/AdminUI/IEI-Dashboard.aspx
NPTEL	http://nptel.ac.in/
National Digital Library	https://ndl.iitkgp.ac.in/

The college main library is also having the following journals subscriptions to support the students and faculty members in research activities.

Sl. No.	Name of the Journal	Frequency
1.	ICI Journal Indian Concrete Institute	Quarterly
2.	Journal of Structural Engineering	Bi-Monthly
3.	Journal of the Institute of Engineers A Series (SCOPUS)	Quarterly
4.	Indian Geotechnical Journal (SCOPUS)	Monthly
5.	NICMAR Journal of Construction Management	Quarterly
6.	Indian Concrete Journal	Monthly

Our Library is extending remote access facility to access multiple e-resources from anywhere and anytime. Access Link: https://idp.ritrjpm.edu.in/

Faculty members motivate the students for using open access journals and make them aware about the latest arrivals.

Department library

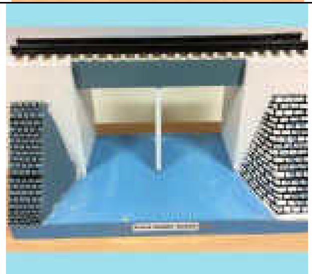
The civil engineering department has its own library, in addition to the main college library, specifically designed to meet the needs of students and faculty. The following materials are available in the civil engineering department library. It also offers books, e-books, journals, e-journals, project reports and video contents.

Particulars	Count (nos.)
Text books	699
Code books	1082 (Title 75)
E-Books	10
E-Journals	192
Project Reports	273
Industrial Training Report	110
Survey Camp Report	53

(D) Working Models

Faculty use models as teaching aids to facilitate better understanding and also encourage the students to create working or display models. Working Models are developed by the faculty and demonstrated to the students for their easy understanding. The Civil Engineering Department has developed the following working models for teaching the subjects Design of Reinforced Concrete Elements & Masonry Structures, Design of Steel Structural Elements & Irrigation and Environmental Engineering Drawing.

Table 5.22 Working Models

Name of the Model	Model Image	Remarks/Specifications
RCC Cantilever Retaining Wall		Length = 2’6” Breadth = 2’ Height=3’
RCC Counter Fort Retaining Wall		Length = 2’6” Breadth = 2’ Height=3’
RCC Tee Beam Bridge		Length = 3’2” Breadth = 1’3” Height=1’3”
Plate Girder Bridge		Length = 3’2” Breadth = 1’3” Height=1’3” Track width = 2”

Name of the Model	Model Image	Remarks/Specifications
Circular RCC Water Tank		Length = 2’6” Breadth = 1’9” Height=1’2” Two one compartment = 1’x1’3”
Rectangular RCC Water Tank		Length = 3’2” Breadth = 1’3” Height=1’3”
Truss Girder Bridge		Length = 2’11” Breadth = 2’3” Height=1’ Truss Height = 9”
Solid Slab Bridge		Length = 3’ Breadth = 1’3” Height=1’6” Road width c-c = 9” Outer = 1’3” Under Road = 8”x1’3”

Name of the Model	Model Image	Remarks/Specifications
Hemispherical Bottomed Steel Tank		Diameter = 1’2” Height = 2’
Aqueduct		System that carry water from a source to a distribution point
Canal Regulator		Structure that controls the flow of water and sediment into the canal
Tank Weir		Specialized equipment designed for effective liquid management

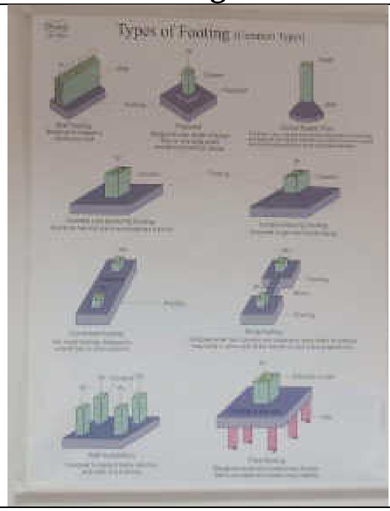
Name of the Model	Model Image	Remarks/Specifications
Flocculator		A device that mixes and retains water to coagulate and flocculate solids in wastewater or surface water
Trickling Filter		A wastewater treatment system that uses a fixed bed of media to degrade organic compounds in liquid waste
Tank Sluice with Tower Head		A large valve in which a rectangular or circular gate slides across the opening
Sluice Gate		A movable barrier that controls the flow of water through a channel, river, canal, or dam

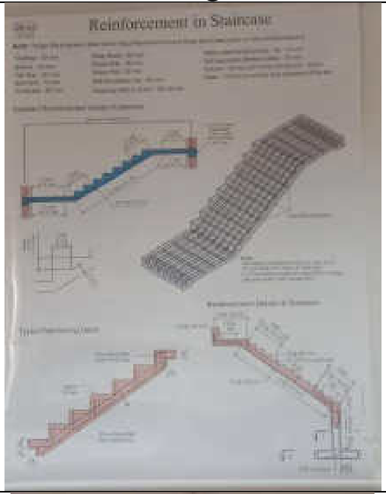
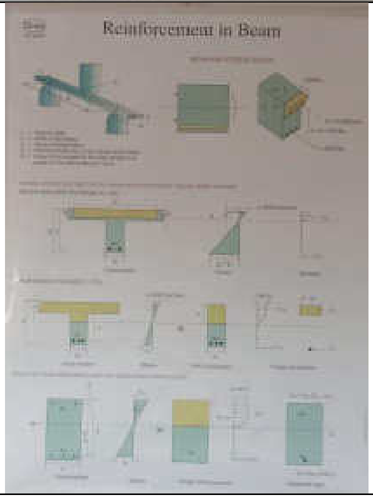
Name of the Model	Model Image	Remarks/Specifications
Septic Tank		An underground chamber made of concrete, fiberglass, or plastic through which domestic wastewater (sewage) flows for basic sewage treatment

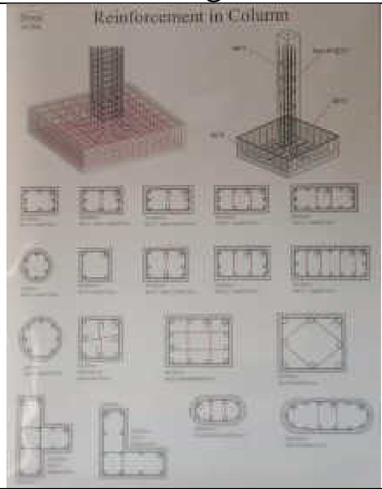
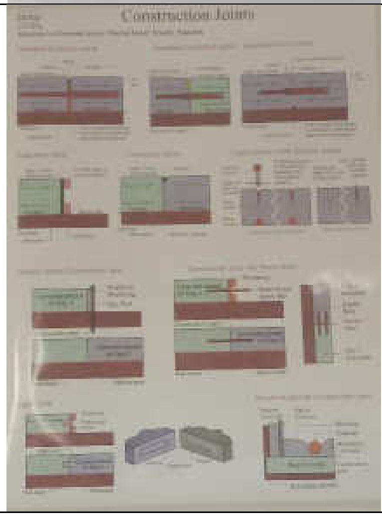
Charts

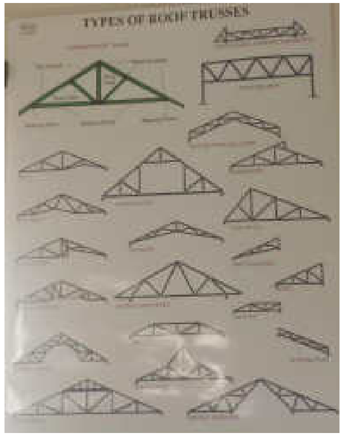
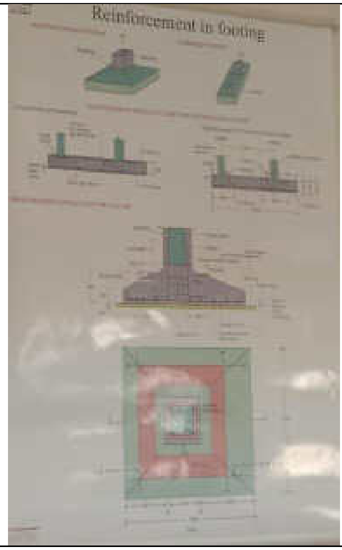
Teaching the contents through charts in the classroom/laboratories made the interaction between students and teachers more effective. The charts available in the department are shown in the table below 5.23

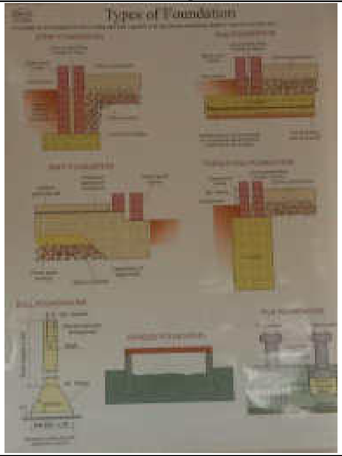
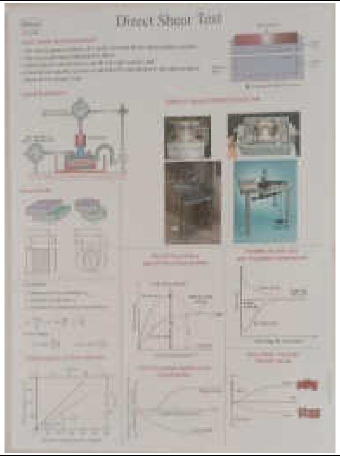
Table 5.23 Charts

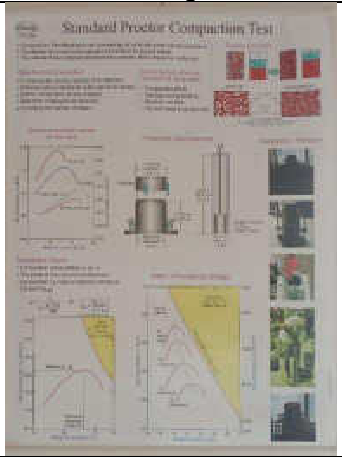
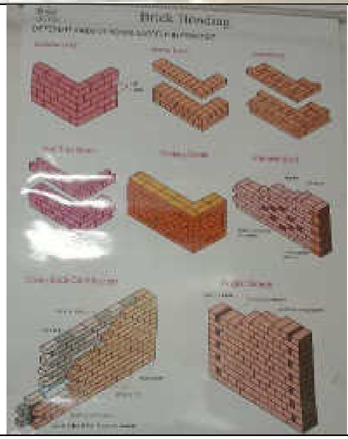
Concept	Image	Remarks
Types of Footing		A chart showing the various types of footing used in construction

Concept	Image	Remarks
Staircase Reinforcement		A pictorial representation of staircase detailing along with its dimensions
Beam Reinforcement		The chart displaying reinforcement details of beam under various loading condition

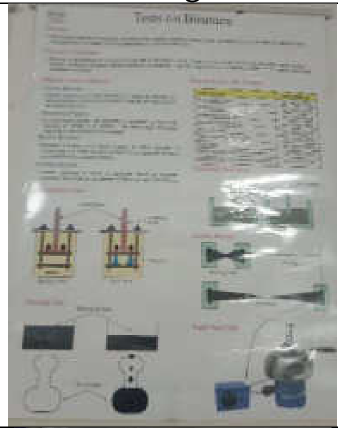
Concept	Image	Remarks
Column Reinforcement		The chart showing various types of columns reinforcement details
Construction Joints		Different types of joints in concrete construction

Concept	Image	Remarks
Types of Roof Trusses		Different types of roof trusses in building structures
Reinforcement in Footing		Reinforcement detailing of various types of footing

Concept	Image	Remarks
Types of Foundation		Various types of foundation used in construction
Direct Shear Test		Procedure for determining the shear strength of soil materials by Direct shear test

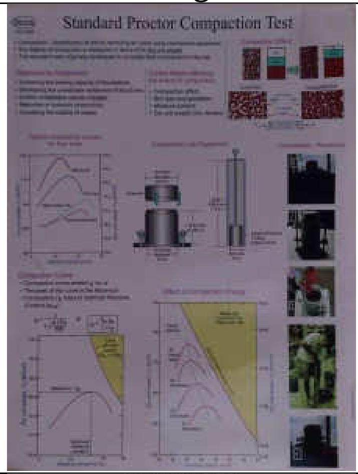
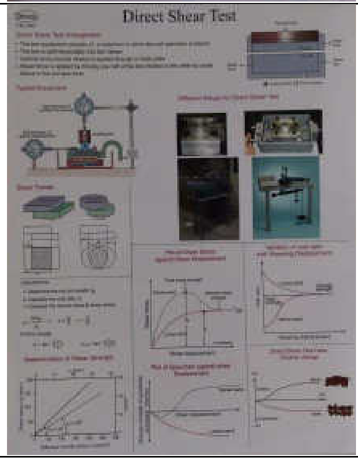
Concept	Image	Remarks
Standard Proctor Compaction Test		Procedure for determining optimum moisture content soils by Standard Proctor Compaction Test
Reference Code Books for Experiments		List of Indian Standard Codal Provisions for the tests related to Soil Mechanics laboratory are shown
Brick Bonding		A pictorial representation of the various types of brick bonding has been helpful in studying the subject construction materials

Concept	Image	Remarks
Types of Flooring		A pictorial representation of the various types of Flooring with its pros and cons has been helpful in studying the subject CE construction materials
CBR		Procedure of CBR Value test are shown for evaluating the strength of subgrade soil and base course

Concept	**Image**	**Remarks**
Tests on Bitumen		The various test on bitumen are displayed to determine the bitumen properties
Flakiness and Elongation Index		Flakiness and Elongation Index test procedure is shown to determine Particle shape of the aggregate specimen

Concept	Image	Remarks
Door and its types		A pictorial representation of the various types of Doors and its specification
Types of Windows		A pictorial representation of the various types of Windows and its specification

Concept	Image	Remarks
Theodolite and Levelling Instruments		A pictorial representation details about the instruments such as Theodolite, Automatic Level and Laser Level
Total Station Instrument		Principle of Total Station and its parts are shown

Concept	Image	Remarks
Standard Proctor Compaction Test		Theory and Principle behind the Standard Proctor Compaction Test is shown
Direct Shear Test		Theory and Principle behind the Direct Shear Test is shown

Concept	Image	Remarks
Reference Code Books for Experiments		List of Indian Standard Codal Provisions for the tests related to Water and Wastewater Analysis laboratory are shown
Reference Code Books for Experiments		List of Indian Standard Codal Provisions for the tests related to ensure the quality of construction materials

Monograms

Monograms are study materials that consist of a collection of formulas used to solve problems. They are useful for students to recall important information while appearing for End Semester University Examinations and competitive examinations. In this way, some monograms are prepared by the faculty members in the Department of Civil Engineering and it is verified by the Head of the Department. The details of the same are shown in the table 5.24

Table 5.24 Monograms

S. No.	Title	Prepared by
1.	CE3301 Fluid Mechanics	Mrs.B.Bharani Baanu
2.	CE3503 Foundation Engineering	Mr.V.Jeevanantham

5.7.4 Consultancy (from Industry) (5)

2023-2024 (CAYm1)

Project Title	Duration	Funding Agency	Amount
Testing on Concrete Cubes	28.08.2023	SLN INFRA CONMIX PVT LTD	750.00
Testing on Concrete Cubes	28.08.2023	ICON READYMIX CONCRETE	750.00
Testing on Concrete Cubes	18.10.2023	SLN INFRA CONMIX PVT LTD	1,500.00
Testing on Concrete Cubes	18.10.2023	SLN INFRA CONMIX PVT LTD	
Testing on Concrete Cubes	18.10.2023	Mr.Mahadeer Ismail	750.00
Testing on Concrete Cubes	19.10.2023	SLN INFRA CONMIX PVT LTD	750.00
Testing on Concrete Cubes	31.10.2023	SLN INFRA CONMIX PVT LTD	750.00
Testing on Concrete Cubes	01.11.2023	SLN INFRA CONMIX PVT LTD	750.00
Testing on Concrete Cubes	08.11.2023	Mr.Mahadeer Ismail	750.00
Testing on Concrete Cubes	08.11.2023	SLN Infra conmix pvt ltd	750.00
Testing on Concrete Cubes	09.11.2023	SLN Infra conmix pvt ltd	750.00
Testing on Concrete Cubes	28.11.2023	Icon Ready Mix Concrete	2,250.00
Testing on Concrete Cubes	28.11.2023	Icon Ready Mix Concrete	
Testing on Concrete Cubes	28.11.2023	Icon Ready Mix Concrete	
Test on water sample	21.12.2023	M/s.Ananya Construction,srivilliputtur,	1,750.00
Testing on Concrete Cubes	22.12.2023	Karupiah site(SLN Infra conmix pvt ltd)	750.00
Testing on Concrete Cubes	29.12.2023	M/s.Shreevenkateshwaraa constructions	1,500.00
Testing on Concrete Cubes	29.12.2023	M/s.Shreevenkateshwaraa constructions	
Test on water sample	30.12.2023	M/s.Ananya Construction,srivilliputtur,	1,750.00
Testing on Concrete Cubes	01.02.2024	M/s.Shreevenkateshwaraa constructions	500.00
Testing on Concrete Cubes	12.01.2024	Er.Karuppaiah	750.00
Testing on Concrete Cubes	22.01.2024	M/s.Shreevenkateshwaraa constructions	1,500.00
Testing on Concrete Cubes	22.01.2024	M/s.Shreevenkateshwaraa constructions	
Field Density Test	20.01.2024	Mr.T.Raja,DCE.,RJ Builders, Rajapalayam	4,000.00
Field Density Test	29.01.2024	Mr.T.Raja,DCE.,RJ Builders, Rajapalayam	3,000.00
Testing on Concrete Cubes	29.01.2024	M/s.Shreevenkateshwaraa constructions	750.00
Testing on Concrete Cubes	01.02.2024	Government Hospital,Rajapalayam	750.00
Test on Interlocking Bricks	26.02.2024	BRICKSA	1,500.00

Project Title	Duration	Funding Agency	Amount
Testing on Concrete Cubes	02.02.2024	Icon Ready Mix Concrete	750.00
Testing on Concrete Cubes	06.02.2024	Icon Ready Mix Concrete	750.00
Testing on Concrete Cubes	18.01.2024	Shree venkateshwaraa construction	500.00
Testing on Concrete Cubes	10.02.2024	Er.Karppaiah	750.00
Testing on Concrete Cubes	16.02.2024	M/S.Icon Ready Mix Concrete	750.00
Testing on Concrete Cubes	24.02.2024	Er.Andrew Celestine Paul	750.00
Testing on Concrete Cubes	24.02.2024	Er.Ganesan, srivillputhur,	1,500.00
Testing on Concrete Cubes	24.02.2024	Er.Ganesan, srivillputhur,	
Testing on Concrete Cubes	22.02.2024	M/s.Shreevenkateshwaraa constructions	500.00
Testing on Concrete Cubes	12.03.2024	M/s.Shreevenkateshwaraa Constructions	750.00
Testing on Concrete Cubes	12.03.2024	Icon Ready Mix Concrete	3,000.00
Testing on Concrete Cubes	19.03.2024	Icon Ready Mix Concrete	
Testing on Concrete Cubes	16.03.2024	SLN INFRA	
Testing on Concrete Cubes	16.03.2024	Icon Ready Mix Concrete	
Testing on Concrete Cubes	16.03.2024	M/s.Shree Siva Builders	1,500.00
Testing on Concrete Cubes	15.03.2024	M/s.Shreevenkateshwaraa Constructions	
Testing on Concrete Cubes	21.03.2024	Icon Ready Mix Concrete	750.00
Standard penetration test	21.03.2024	Mrs.Shyamala	5,000.00
Testing on Concrete Cubes	22.03.2024	Icon Ready Mix Concrete	750.00
Testing on Concrete Cubes	27.03.2024	M/s.Shree Siva Builders	1,500.00
Testing on Concrete Cubes	04.04.2024	Shree Siva Builders(ANJAC)	
Testing on Concrete Cubes	05.04.2024	Shree Siva Builders(ANJAC),new sf block	3,000.00
Testing on Concrete Cubes	01.04.2024	Icon Ready Mix Concrete	
Testing on Concrete Cubes	03.04.2024	Shree Siva Builders(ANJAC),new sf block	
Testing on Concrete Cubes	05.04.2024	Shree Siva Builders(ANJAC),new sf block	
Testing on Concrete Cubes	05.04.2024	Shree Siva Builders(ANJAC),new sf block	750.00
Testing on Concrete Cubes	25.04.2024	M/S.Icon Ready Mix Concrete	1,500.00
Testing on Concrete Cubes	27.04.2024	M/s.Shree Siva Builders	
Testing on Concrete Cubes	18.04.2024	M/s.Shree Siva Builders	1,500.00
Testing on Concrete Cubes	27.04.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	29.04.2024	M/s SLN Infra conmix private Limited	750.00

Project Title	Duration	Funding Agency	Amount
Testing on Concrete Cubes	27.04.2024	M/s.Icon Ready Mix Concrete	3,000.00
Testing on Concrete Cubes	29.04.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	25.04.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	27.04.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	27.04.2024	M/s.Icon Ready Mix Concrete	6,750.00
Testing on Concrete Cubes	09.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	09.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	10.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	10.05.2024	M/s.Icon Ready Mix Concrete	
Test on Interlocking Bricks	10.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	10.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	14.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	11.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	14.05.2024	M/s SLN Infra conmix pvt.Ltd	750.00
Testing on Concrete Cubes	20.05.2024	M/s.Icon Ready Mix Concrete	3,000.00
Testing on Concrete Cubes	20.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	20.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	20.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	6,750.00
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	31.05.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	01.06.2024	M/s.Icon Ready Mix Concrete	1500.00
Testing on Concrete Cubes	01.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	05.06.2024	M/s.Icon Ready Mix Concrete	750.00
Testing on Concrete Cubes	11.06.2024	Aruna multi Hospital	2,250.00
Testing on Concrete Cubes	11.06.2024	Aruna multi Hspital	

Project Title	Duration	Funding Agency	Amount
Testing on Concrete Cubes	07.06.2024	Aruna multi Hspital	
Testing on Concrete Cubes	10.06.2024	M/s.Icon Ready Mix Concrete	2,250.00
Testing on Concrete Cubes	10.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	10.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	11.06.2024	SLV Ready mix concrete	750.00
Testing on Concrete Cubes	15.06.2024	M/s.Icon Ready Mix Concrete	1,500.00
Testing on Concrete Cubes	15.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	15.06.2024	SLVReady Mix Concrete	2,250.00
Testing on Concrete Cubes	15.06.2024	SLVReady Mix Concrete	
Testing on Concrete Cubes	15.06.2024	SLVReady Mix Concrete	
Testing on Concrete Cubes	20.06.2024	M/s.Icon Ready Mix Concrete	1,500.00
Testing on Concrete Cubes	20.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	15.06.2024	M/s.Icon Ready Mix Concrete	1300.00
Testing on Concrete Cubes	15.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	15.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	15.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	25.06.2024	M/s.Icon Ready Mix Concrete	1500.00
Testing on Concrete Cubes	25.06.2024	M/s.Icon Ready Mix Concrete	
Testing on Concrete Cubes	25.06.2024	M/s.SLN Infra concrete mix	3000.00
Testing on Concrete Cubes	25.06.2024	M/s.SLN Infra concrete mix	
Testing on Concrete Cubes	25.06.2024	M/s.SLN Infra concrete mix	
Testing on Concrete Cubes	25.06.2024	M/s.SLN Infra concrete mix	
Total			**94,800.00**

2022-2023 (CAYm2)

Project Title	Duration	Funding Agency	Amount
Test on water	22.07.2022	DAVINCH Builders, Rajapalayam.	150.00
Field Density Test	02.08.2022	Sabari Construction	4,000.00
Standard penetration Test	28.09.2022	Archana Blocks	12,500.00
Survey works	28.09.2022	Archana Blocks	

Project Title	Duration	Funding Agency	Amount
Testing of Bricks	14.10.2022	SRJ & Chola packaging pvt.,Ltd	1,500.00
Field Density Test	28.11.2022	RML Fabrics Division	3,000.00
Testing of Bricks	14.02.2023	KVP Flyash Bricks	3,250.00
Testing of Bricks	16.02.2023	Sree Kumaran Flyash Bricks	2,500.00
Test on water sample	22.02.2023	Sam Builders	1,750.00
Test on water sample	23.02.2023	Sam Builders	1,750.00
Test on water sample	10.03.2023	Sam Builders	1,750.00
Test on Mortar cube & cylinder	16.03.2023	Mr.E.vipurajan & J.N.Puvaneshwaran	2,500.00
Test on water sample	13.03.2023	Mrs.Premalatha (Research scholar)	150.00
Test on Concrete cube & cylinder	21.03.2023	Mrs.Premalatha (Research scholar)	1,500.00
Total			**36,300.00**

2021-2022 (CAYm3)

<table>
<tr><th>Project Title</th><th>Duration</th><th>Funding Agency</th><th>Amount</th></tr>
<tr><td>Testing on Bricks</td><td>17.08.2021</td><td>Ayyan Bricks</td><td>1,000.00</td></tr>
<tr><td>Field Density Test</td><td>18.08.2021</td><td>RVP Builders</td><td>11,000.00</td></tr>
<tr><td>Field Density Test</td><td>25.08.2021</td><td>RVP Builders</td><td>4000.00</td></tr>
<tr><td>Field Density Test</td><td>27.08.2021</td><td>RVP Builders</td><td>2,000.00</td></tr>
<tr><td>Testing onBricks</td><td>31.08.2021</td><td>Darling Bricks</td><td>2,250.00</td></tr>
<tr><td>Field Density Test</td><td>01.09.2021</td><td>RVP Builders</td><td rowspan="10">24,000.00</td></tr>
<tr><td>Field Density Test</td><td>01.09.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>02.09.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>14.09.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>13.09.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>14.09.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>28.09.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>29.09.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>14.09.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>09.10.2021</td><td>RVP Builders</td></tr>
<tr><td>Field Density Test</td><td>09.10.2021</td><td>RVP Builders</td><td rowspan="2">29,000.00</td></tr>
<tr><td>Field Density Test</td><td>16.10.2021</td><td>RVP Builders</td></tr>
</table>

Project Title	Duration	Funding Agency	Amount
Field Density Test	29.10.2021	RVP Builders	
Field Density Test	01.11.2021	RVP Builders	
Field Density Test	02.11.2021	RVP Builders	
Field Density Test	19.11.2021	RVP Builders	8,000.00
Field Density Test	19.11.2021	RVP Builders	
Field Density Test	24.11.2021	RVP Builders	
Field Density Test	24.11.2021	RVP Builders	
Field Density Test	20.12.2021	RPS & Co Southern Railway	10,000.00
Field Density Test	31.12.2021	RPS & Co Southern Railway	10,000.00
Standared penetration Test	05.01.2022	ESSARR Associates	18,000.00
Testing on Concrerte cube	07.02.2022	Thenmozhi (Research scholar)	500.00
Standared penetration Test	09.02.2022	Chitra multi- specialty hospital	18,000.00
Test on sand	30.03.2022	Rajapalayam Municipality	1,000.00
Test on water sample	21.04.2022	Government Hospital,Rajapalayam	4,100.00
Field Density Test	12.05.2022	RVP Builders	2,000.00
Field Density Test	08.06.2022	RVP Builders	3,000.00
Field Density Test	13.06.2022	RVP Builders	6,000.00
Field Density Test	30.06.2022	Sabari constructin	3,000.00
Field Density Test	30.06.2022	Sabari constructin	3,000.00
		Total	**1,59,850.00**

5.8. Faculty Performance Appraisal and Development System (FPADS) (30)

The Ramco Institute of Technology have a well-defined and transparent Faculty Performance Appraisal and Development System. The faculty members asked to submit the well-structured self-appraisal form for each academic year for evaluation by Head of the Department, Principal and Governing Council members. Eligible faculty members are asked to submit the Application for promotion under career advancement scheme along with the self-appraisal form. The newly recruited faculty members complete their probation after one year. The faculty members will get eligible for promotion/career advancement after attaining the required norms. The performance of each and every faculty member will be evaluated every year through a centralized appraisal system, apart from the feedback received from students for each semester.

The main highlights of the appraisal system are listed below

Table 5.25 Evaluation criterion and indices for faculty performance appraisal and development

S.No.	Criteria for Evaluation	Indices
1.	Teaching, Learning and Evaluation related activities	1. Innovations/Contributions in Teaching 2. Teaching performance 3. CO Attainment 4. Supervisory support and Project guidance provided 5. Activities participated (Seminars/Webina Workshops/ FDPs/STTPs) 6. Online Course Details 7. Additional Certification Course 8. Contribution towards Identification of curricul gap/ addressed by content beyond syllabus (Theory/practical) 9. Activities that contribute to student success in the form of improved and Measurable learning outcomes. 10. Mentoring effectiveness
2.	Research, Publications and Academic contributions	1. Ph.D. Guidance 2. Papers presented in International / National Conference 3. Research Papers published in National/ International Journal 4. Research Papers Communicated to National/ International Journal 5. Book/Book Chapter published, Monographs, Lab Manuals authored 6. Research Grants/ Funded Projects applied 7. Research Grants/ Funded Projects obtained

S.No.	Criteria for Evaluation	Indices
		8. IP Rights filed / granted 9. Research related service (Reviewing journals, editorial roles, organizing research seminars, conferences, etc.) 10. Contribution to industrial interactions in the form of consultancy/ sponsored R&D
3.	Career Enhancement/ Extension activities	1. Contribution in organizing seminar/webinar/ workshops/ conferences/ symposium/ FDPs/STTPs/ other programme 2. Participation as Resource Person for delivery of special lectures / Chairing sessions/ Jury etc. 3. Contribution to Society 4. Achievements / Awards received during the period 5. Recognition received during the period 6. Initiation of MoU with reputed universities/ research centres/organizations 7. Contribution to improvement in Training & Placement/Career Guidance cell/ EDC/ IIC etc.
4.	Administration	1. Institutional Development Elements: College level / Department Level 2. Industry/ Institute Contribution: Effectiveness of MOU/ Industrial Collaboration Outcome 3. Activities that support institute accreditation/ other development contributions 4. Interaction outside RIT
5.	Evaluation of Faculty Performance by the HoD	Assessment of Faculty Members by the HOD and Assessment of HOD by the Principal

1.Sample of Annual Performance Appraisal form:

RAMCO INSTITUTE OF TECHNOLOGY
Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Annual Performance Appraisal & Training Identification for

FACULTY MEMBER

Academic Year: 20… - 20…

FORM – A

(to be filled in by the Faculty Member)

Name of the Faculty Member :
Faculty ID :
Date of Birth :
Designation :
Highest Qualification :
Department / Centre :
Date of Joining the Institute :
Present post held from :
Period of Appraisal : ……………… to ………………

Subjects Handling (Odd/Even Semester):

Sl. No.	Semester	Course Code	Name of the Course	Class Strength
1.				
2.				

1. Teaching and Learning:

1.1. Innovations/ Contributions in Teaching: **(20 Marks)**

Course code	Particulars	Details
	Theory Course	
	Usage of NPTEL/ SWAYAM/ Online course Material	
	LMS	
	Use of ICT Tools	
	e- learning content developed/ created with reference details	
	Assessment methods Used	
	Formulation of mathematical model & solving using Simulation tool / softwares	
	Innovative Practices Employed	

Course code	Particulars	Details
	Laboratory Course	
	New experiment created in Laboratory	
	Additional experiments employed	
	Virtual Lab Experiments used	
	Assessment Methods Used	
	Innovative Practices Employed	

1.2. Teaching performance (Result produced during Academic year 20… - 20… Even & 20… - 20… Odd semester): **(20 Marks)**

Sl. No.	Semester	Course Code	Name of the Course taught	Class strength	Result produced (Pass %)
1.					
2.					

1.3. CO Attainment (Academic year 20… - 20… Even & 20… - 20… Odd semester): **(10 Marks)**

Sl. No.	Course Code/ Title	CO1	CO2	CO3	CO4	CO5	Over all CO attainment
1.							
2.							

1.4. Supervisory support and Project guidance provided: Projects guided at B.E./B.Tech. (5 marks)

Sl. No.	Title of the Thesis / Project	Name(s) of the Student(s)	Status (completed / in progress)	Tangible Outcome
1.				
2.				

1.5 (a). Activities participated (Seminars/Webinars): **(2 marks)**

Sl. No.	Title of the Programme	Duration	Organizing Institute	In collaboration with	Level (International/ National)	Outcome
1.						
2.						

1.5 (b). Activities participated (Workshops): **(3 marks)**

Sl. No.	Title of the Programme	Duration	Organizing Institute	In collaboration with	Level (International/ National)	Outcome
1						
2						

1.5 (c). Activities participated (FDPs/STTPs): **(5 marks)**

Sl. No.	Title of the Programme	Date & Duration	Organizing Institute	AICTE/ ATAL/ NITTTR etc.	Level (International/ National)	Outcome
1.						
2.						

1.6. Online Course Details: **(10 marks)**

Sl. No.	Name of the Course	University	Platform (EDX/ Coursera)	Duration/ Completed on	Outcome
1.					
2.					

1.7. Additional Certification Course: **(5 marks)**

Sl. No.	Name of the Course	Organized by	Duration/ Completed on	Outcome
1.				
2.				

1.8 Contribution towards Identification of curricular gap/ addressed by content beyond syllabus etc. (Theory/practical) **(10 marks)**

....

....

1.9 Activities that contribute to student success in the form of improved and measurable learning outcomes **(5 marks)**

....

....

1.10 Mentoring effectiveness **(5 marks)**

....

....

2. Research and Development:

2.1. Ph.D. Guidance

Sl. No.	Title of the Thesis	Name(s) of the Student(s)	Status (completed / in progress)
1.			
2.			

2.2. Papers presented in National Conference: (Refereed conference Paper - Published during the period) * **(10 marks)**

Information to be given in the order as below:

Author Names (Sequence as in paper), Title of the paper, Name of the conference, place, year, Page No.

2.3. Papers presented in International Conference: (Refereed conference Paper - Published during the period) * **(15 marks)**

Information to be given in the order as below:

Author Names (Sequence as in paper), Title of the paper, Name of the conference, place, year, Page No.

2.4. a. Research Papers published in National/ International Journal: (Refereed Journal Paper - Published during the period) * **(20 marks)**

Information to be given in the order as below:

Author Names (Sequence as in paper), Title of the paper, Name of the Journal, Vol. No. (year), Page No.

* Papers only in Vidwan Portal and Website will be considered.

2.4. b. Research Papers Communicated to National/ International Journal: (Refereed Journal Paper - Published during the period) * **(10 marks)**

Information to be given in the order as below:

Author Names (Sequence as in paper), Title of the paper, Name of the Journal

2.5. Book/Book Chapter published, Monographs, Lab Manuals authored: **(10 marks)**

Information to be given in the order as below:

Author Names (Sequence as in paper), Title of the Book, Name of the Publisher, Edition and year of Publication.

2.6. (a) Research Grants/ Funded Projects applied: **(5 marks)**

Sl. No.	Title of the Research Project	Grant applied (Rs. in Lakhs)	Name of the Funding Agency	Project Duration (Years)
1.				

2.6. (b) Research Grants/ Funded Projects obtained: **(10 marks)**

Sl. No.	Title of the Research Project	Grant received (Rs. in Lakhs)	Name of the Funding Agency	Project Duration (Years)
1.				

2.7. IP Rights filed / granted: **Yes / No** **(5 marks)**

Sl. No.	Title	Application No.	Filed Date	Status	Inventor(s)
1.					

2.8. Research related service (Reviewing journals, editorial roles, organizing research seminars, conferences, etc.) **(5 marks)**

………………………………………………………………………………………………………

………………………………………………………………………………………………………

2.9. Contribution to industrial interactions in the form of consultancy/ sponsored R&D **(10 marks)**

………………………………………………………………………………………………………

………………………………………………………………………………………………………

3. Career Enhancement/ Extension activities

3.1 (a). Contribution in organizing seminar/webinar: **(10 marks)**

Sl. No.	Title of the Programme	Date & Duration	Sponsor(s)	Level (International/ National)	Nature of Contribution	Beneficiaries
1.						
2.						
3.						

3.1 (b). Contribution in organizing workshops: **(10 marks)**

Sl. No.	Title of the Programme	Date & Duration	Sponsor(s)	Level (International/ National)	Nature of Contribution	Beneficiaries
1.						
2.						
3.						

3.1 (c). Contribution in organizing conferences: **(10 marks)**

Sl. No.	Title of the Programme	Date & Duration	Sponsor(s)	Level (International/ National)	Nature of Contribution	Beneficiaries
1.						
2.						
3.						

3.1 (d). Contribution in organizing symposium: **(10 marks)**

Sl. No.	Title of the Programme	Date & Duration	Sponsor(s)	Level (International/ National)	Nature of Contribution	Beneficiaries
1.						
2.						
3.						

3.1 (e). Contribution in organizing FDPs/STTPs: **(10 marks)**

Sl. No.	Title of the Programme	Date & Duration	Sponsor(s)	Level (International/ National)	Nature of Contribution	Beneficiaries
1.						
2.						

3.1 (f). Contribution in organizing other programme (which are not listed in 3.1 (a) to 3.1 (e)): **(10 marks)**

Sl. No.	Title of the Programme	Date & Duration	Sponsor(s)	Level (International/ National)	Nature of Contribution	Beneficiaries
1.						
2.						

3.2. Participation as Resource Person for delivery of special lectures / Chairing sessions/ Jury etc. **(10 marks)**

Sl. No.	Nature of the Programme	Title of the Topic Delivered	Date(s)	Organization	Level (International/ National)
Internal					
1.					
2.					
3.					
External					
1.					
2.					

3.3. Contribution to Society: **(5 marks)**

-
-

3.4. Achievements / Awards received during the period (please specify): **(5 marks)**

3.5. Recognition received during the period (please specify): **(5 marks)**

3.6. Initiation of MoU with reputed universities/ research centres/organizations (5 marks)

....

....

3.7. Contribution to improvement in Training & Placement/Career Guidance cell/ EDC/ IIC etc. (10 marks)

....

....

4. Administration

4.1. Institutional Development Elements:

a. College Level (20 marks)

Sl.No.	Position held	Period	Responsibility	Activities carried out
1.				•
2.				•
3.				•

b. Department Level (30 marks)

Sl.No.	Position held	Period	Responsibility	Activities carried out
1.				•
2.				•
3.				•

4.2. Industry/ Institute Contribution

a. Effectiveness of MOU: (20 marks)

Sl. No.	Name of the Industry	Outcome
1.		
2.		

b. Industrial Collaboration Outcome: (10 marks)

Sl. No.	Name of the Industry	Role	Outcome
1.			
2.			

4.3. Activities that support institute accreditation/ other development contributions (15 marks)

....

....

4.4. Interaction outside RIT (5 marks)

....

....

Self-Assessment Score:

Criteria	Marks Obtained	Performance score
Teaching and Learning (100)		
Research and Development (100)		
Career enhancement /Extension activities (100)		
Administration (100)		
Total		

[Guideline for Self-Appraisal rating: Excellent: > 90%; Very Good: 80-90%; Good: 70-80%; Average: 60-70%; Poor <60%]

Overall Self–Appraisal rating (please tick):
Poor /Average /Good /Very Good /Excellent

a. Identification of Training needs (please tick the relevant need for improvement):
- Participation in FDP / SDP / STTP in the subject area for training and updating
- Participation and presentation of papers in conferences / seminars / symposia
- To pursue Higher Education for qualification upgrading and enhanced learning
- Any other need (specify)

b. Training need analysis (Summary with reason):

....

....

....

....

c. How much total incentives received from the institute, as appreciating your activities?

....

....

d. State the potential scope for your development in RIT.

....

....

e. **How much concession you received as contribution by welfare scheme of RIT?**
   a. **School Children: Rs.**
   b. **Bus:**
   c. **Lunch:**
   d. **Medical Insurance: Rs.**

f. **Comment on the work including particulars of circumstances for not being able to undertake activities:**

....
....
....
....

g. **Your Plan for next 2 Years:**

....
....
....
....

h. **Comments/ Suggestions for future scope of College/ Department:** (Including difficulties faced if any and suggestions for improvement, training, infrastructure etc. for professional growth and for achievement of excellence)

....
....
....
....

Date: **Name & Signature of the Faculty Member**

Weightage of marks obtained as per the cadre for performance score:

Sl.No.	Module	P	ASCP	AP(SG)	AP
1.	Teaching and Learning (100)	50%	60%	70%	80%
2.	Research and Development (100)	25%	20%	15%	10%
3.	Career Enhancement /Extension activities (100)	10%	10%	7.5%	5%
4.	Administration (100)	15%	10%	7.5%	5%

FORM – B

Assessment of the Faculty Performance

(Assessment of Faculty Members by the HOD and Assessment of HOD by the Principal)

Name of the Faculty Member:

Designation & Department :

a. **Attitude, Interpersonal Skills, Soft Skills and Personality** (give ratings on a 5-point scale)

(5: Excellent; 4: Very Good; 3: Good; 2: Satisfactory; 1: Poor)

1.	Attitude: right thinking, self-disciplined and has unbiased approach	
2.	Initiative: self-starter, able to work with interest and goal-oriented	
3.	Punctuality: arrives on time, available fully during working hours & always engaged	
4.	Responsibility: understands duties, accepts responsibilities readily	
5.	Commitment: committed to his/her work, not taking unnecessary leaves and completes tasks on time	
6.	Loyalty: supports and follows Institute's policies and guidelines	
7.	Professional development: keeps updating knowledge regularly & Does contemporary research	
8.	Innovative: Interested in learning new subjects, fits to online environments and using modern teaching techniques & LMS tools	
9.	Oral communication: speaks effectively in English with students, colleagues and higher authorities	
10.	Written communication: quality of letter drafting, content and modalities	
11.	Leadership and Team work: effective in a team work, takes initiative, gives clear directions, offers guidance to junior faculties and assumes leadership role	
12.	Accountability: accepts his/her role in results and outcomes	
13.	Support: extends support to students, motivates, counsel and mentor for learning, helps students to improve their understanding level	
14.	Ethics and professionalism	
15.	Maturity and Temperament	
16.	Sincerity, decency and dedication at work	
17.	Relationship with students (strictness, guidance, not cheap popularity)	
18.	Relationship with fellow faculty and staff members (team work)	
19.	Relationship with higher Officers and College administration (obedient & sincere to seniors)	
20.	Syllabus completion: only by topic/content or elaboration ie beyond syllabus	
21.	Contribution towards student's skill development: encourages students to do Seminars / Presentations	
22.	Placement support: Ready to take Logical, Aptitude, Reasoning and Programming	
23.	Purchase support: knows the procedure, helps in procurement	
24.	Contribution to Department: novel contributions, develops new experiments, utilizes all the equipment in laboratory, support to Co-curricular and Extra-curricular activities	

25.	Library utilization: Regularly visits library, lends book & make students to visit library	
	Total (out of 125)	
	Overall Rating out of 50	

b. In a year how many days he/she was on leave? Last year?

....

c. Verified that all his/her Claims in form A are correct and genuine? (Yes / No)
Your comments:

....
....
....

d. Comments / Remarks / Observations of the Reporting Officer

HOD:

....
....
....
....
....
....
....

Overall score assigned by the HOD for the individual based on the details provided in PART A: ....

Signature of HOD

e. Comments given by individual & HOD: Review

Principal:

....
....
....
....
....

....
....

Overall score for the individual awarded by the principal based on Part A and Part B:

....

Signature of Principal

Total Score and overall performance rating:

Form-A (100)	Form-B (50)	Students' Feedback (25)	Principal (25)	Total (200)	Percentage (100)	Overall performance rating

[Guideline for overall performance rating: Excellent: > 90%; Very Good: 80-90%; Good: 70-80%; Average: 60-70%; Poor <60%]

Submitted to the Management Committee for assessment:

1. Performance Appraisal conducted on: ....
2. Reviewed the above credentials by: ....
3. Assessment done by the above: Approved/ not Approved
4. Suggestions/ critics for the individual:

....
....
....

Signature:

CEO **Governing Council Member**

FORM C

Status of actions taken on feedback received from previous performance appraisal

SELF ANALYSIS COMMENTS:

Sl. No.	Field of Comment	Actions Taken
1.	Training need analysis	
2.	Potential Scope for development in RIT	
3.	Comment on work including particulars of circumstances for not being able to undertake activities	
4.	Plan for next 2 years	
	Signature of the Faculty member	

COMMENTS BY REPORTING OFFICER (HoD):

Sl. No.	Field of Comment	Actions Taken
1.	Comment given by the reporting officer	
	Signature of the HOD	

COMMENTS BY PRINCIPAL:

Sl. No.	Field of Comment	Actions Taken
1.	Comment given by the Principal	
	Signature of the Principal	

COMMENTS BY MANAGEMENT COMMITTEE:

Sl. No.	Field of Comment	Actions Taken
1.	Comment given by the Management Committee Member	
	Signature of the Management committee Member	

2.Sample of application for promotion under career advancement scheme form:

RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

APPLICATION FOR PROMOTION UNDER CAREER ADVANCEMENT SCHEME

1. Name of the Faculty Member:

2. Promotion applied for:

3. (i) Academic Profile:

Degree	Branch/ Specialization	Name of the Institution	University/ Board	Year of Passing	Percentage/ CGPA &Class Obtained

(ii) Research Degree

Degree	Title of the Thesis	Date of Award as per University notification	Name of the University
M.Phil			
Ph.D			
Post. Doc/D.Sc			
Any other			

4. Experience:

(i) Posts held after appointment to this Institution:

Designation	Department	From	To	Pay band with Grade pay

(ii) Appointments held prior to joining this Institution:

Designation	Name of the Employer	Date of Joining	Date of leaving	Pay band with Grade pay	Reason for leaving

5. Additional Contribution/ Responsibilities in our Institution:

(i) Department Level:

(ii) Institute Level:

6. Detailed list of developmental works/ Contributions to the Department:

7. Brief details of FDTPs/ Summer/ Winter Schools/ Conferences/ Seminars/ Symposia/ Workshops attended (Add extra sheets if necessary)

Sl.No.	Name of Programme/event	Duration		Total number of days
		From	To	

8. Brief details of FDTPs/ Summer/ Winter schools/ Conferences/ Seminars/ Symposia/ Workshops/ value added courses organised (Add extra sheets if necessary)

Sl.No.	Name of Programme/event	Duration		Total number of days
		From	To	

9. Teaching and Learning

(i) Teaching Performance

Academic year	Semester	Course code	Course name	Pass Percentage

(ii) Innovative practices employed in Teaching and assessment, if any:

(iii) List of Online Courses Completed:

(iv) Details of Books/manuals/other teaching materials developed:

10. Details of Professional development activities such as Industrial visits/IPT/membership in professional society, etc.,:

11. Details of Research and Development contributions:

(i) List of papers published in Journals (Attach as annexure):

(ii) List of papers presented in Conference (Attach as annexure):

(iii) List of funded projects/consultancy projects/patents, if any:

(iv) List of projects/Research guidance (Attach as annexure):

12. Other relevant information/Prizes, Medals, Awards received, if any:

Signature of the Faculty Member **Signature of the HOD**

(B) Implementation and Effectiveness:

After obtaining the faculty members self-appraisal forms and career advancement form for promotion, an interview will be arranged and the concerned faculty will be informed.

A scrutiny committee comprising senior professors / other department HoDs will to check and verify details furnished in the self-appraisal forms and promotion form. Further this committee will submit one-page consolidated report to the interview panel

The interview panel, comprising the Chief Educational Officer or Chief Operating Officer of the Trust, a Governing Council Member, the Principal, the Vice-Principal, and the Head of the Department, discusses each faculty member based on the appraisal form and the report received from the scrutiny committee. Based on the overall performance rating, the following decisions are recommended by the panel for further action.

1.Rating with Outstanding, Excellent, Very Good performance are recommended for

- Promotion through career advancement scheme
- Registering for Ph.D.

The following suggestions for improvement are given

- Getting Funding proposals
- Suggestions for getting Awards and achievements
- Promotion through career advancement scheme

Promotion through career advancement scheme

Based on the career advancement scheme, the promotion offered is give in table 5.26

Table 5.26 Details of faculty members promoted through career advancement scheme for the past three assessment years

S. No.	Name of the Faculty Member	Designation	Promoted as	Since from
1.	Dr.S.Dharmar	Assistant Professor (Senior Grade)	Associate Professor	01.01.2023
2.	Dr.M.Indhumathi	Assistant Professor	Associate Professor	01.01.2023
3.	Mr.T.Chockalingam	Assistant Professor	Assistant Professor (Senior Grade)	01.01.2023

4.	Dr.G.Karthikeyan	Assistant Professor (Senior Grade)	Associate Professor	01.02.2024

Faculty members Registering for Ph.D.

Table 5.28 Details of Ph.D. Registration

S. No.	Name of the Faculty	Register No.	Date of Registration
1.	Mrs.A.Leema Margret	23241791141	Jan 2023
2.	Mr.A.Manicka Mamallan	22141797115	July 2022
3.	Mr.R.Muruganantham	22141797113	July 2022

2.Rating with Good and Average performance the following suggestions for improvement is given

- Identification of training needs
- Presenting and publishing Papers in Conference and Journals

Based upon the recommendation given by the committee, the need for various training programme or higher qualification requirements are identified. The HODs review these with the individual training records. These are forwarded to the Principal with their recommendation. Based on this, the desired training, or skill enhancement opportunities are provided to the faculty members to enhance their knowledge in the emerging technology.

3.Rating with satisfactory and poor

The faculty members with satisfactory and poor rating will be given a chance for improvement and if the same scenario is continued further actions are taken.

5.9 Visiting/Adjunct/Emeritus Faculty etc. (10)

The department and institute have realized the need for the provision of appointing adjunct faculty in order to enhance the efficiency of the department. In this regard, the department has appointed Er. B. Muthukumarasamy, Structural consultant & contractor, Absara Consultancy, Rajapalayam as **adjunct faculty**. In addition, **few more members visited** the department and handled training classes for the students of the Civil Engineering department for the past three academic years. Moreover, the department have **honorary professor** Dr. Mahmoud Nawaf Al-Khazaleh, Assistant Dean of Scientific Research, Department of Civil Engineering, Munib and Angela Masri Faculty of Engineering, University of Technology, Aqaba-Jordan, to enhance Academic and Research collaboration.

Table 5.29 Details of content delivery by Adjunct / Visiting Faculty

S. No.	Handled by	Course Title	Date	No. of hours handled	Content Delivered

1	Er. B. Muthukumarasamy, Proprietor, Absara Consultancy, Rajapalayam	CE3501 Design of Reinforced Concrete Structural Elements	24.10.2024	8 hours	Design of different types of Columns as per IS codal provisions
2.	Mr. K. Iyappan, Project Lead, Tech Mahindra India Pvt. Ltd., Bengaluru	Hands on Training on Project Planning and Management using Primavera and MS Project	25.10.2024 to 26.10.2024	16 hours	Project Planning, Network diagram, Project Scheduling, Roles & Resources, and Report Preparation in PPM and Hands-on Training in Primavera and MS Project.
3	Mr. M.Senthil Murugan, Founder & CEO, BEYCAN Technical Training Institute, Rajapalayam	Internet of Things (IoT) in Civil Engineering	22.08.2023	2 Hours	Importance of (IoT) in Civil Engineering
4	Mr. R. Ganesh Kumar, Senior Structural Consultant TechAppss Consulting, Chennai	Two-day Hands-on Training on STAAD.Pro Connect Edition V22	27.06.2023 to 28.06.2023	2 Days (16 Hours)	Structural Analysis and Design
5	Ms.R.Abinaya, Tekla Modeller, Struct Mech Engineers Pvt. Ltd., Bangalore Alumni – Batch 2016-2020	An overview on TEKLA Software	26.04.2023	3 Hours	1.Structural Design and Detailing 2.Collaboration and Coordination 3. Enhancing Accuracy and Reducing Errors. 4. Support for Precast, Steel, and Concrete Design.
6	Er.M.Ponkumaran, Pilot Drone Trainer Alumni – Batch 2015-2019	Third Eye in the sky	04.03.2023	3 Hours	Drone Surveying
7	Mr. R. Ganesh Kumar, Senior Structural Consultant TechAppss Consulting, Chennai	Hands-on Training on STAAD.Pro Connect Edition V22	27.01.2023 to 28.01.2023	16 Hours	Structural Analysis and Design

8	Mr. Akbar Ali Khader, Senior Application Engineer TechApps Consulting, Chennai	Value added course on Open Building Designer	25.08.2022 to 26.08.2022	2 Days (16 Hours)	BIM workflows to provide information-rich models for the design, analysis, simulation, and documentation of buildings.
9	Mr. K. Iyappan, Project Lead, IBM India Pvt Ltd, Bangalore	Project Planning and Management using Primavera (P6)	25.07.2022 to 30.07.2022	6 Days (48 Hours)	Project planning, scheduling and resource management to budgeting and risk management.
10	Mr.S.Rajesh Kumar, Director, Technical services, TechApps Consulting Services, Chennai	Standard Operating Procedure for Bentley Certification	19.05.2022	3 Hours	Delivered Preparedness for the Certification Exam
11	Mr.S.Rajesh Kumar, Director, Technical services, TechApps Consulting Services, Chennai	Reality Modeling	18.05.2022	3 Hours	Delivered detailed, accurate digital representations of existing structures or landscapes, enabling stakeholders to visualize the environment in 3D.
12	Chandra Shekar, Co-Founder of TurnBIM Engineering Services	Big Benefits of Building Information Modelling (BIM)	22.04.2022	3 Hours	Building Information Modelling
13	Er. S.Suganya, Infinity PMC Solutions Pvt. Ltd., Chennai	Value Added Course on Project Planning and Management using Primavera (P6)	11.11.2021 to 15.11.2021	5 days (40 hours)	Project planning, scheduling and resource management to budgeting and risk management.
14	Dr. Mahmoud Khazalah Assistant Professor in Civil Engineering, Munib & Angela Masri Faculty of Engineering, Aqaba University of Technology, Jordan.	International Conference on Smart Technologies and Applications (ICSTA 2022)	12.03.2022	2 Hours	Invited Talk on Soil Reinforcement in Road Construction Using Geogrids & Geosynthetics Material

15	Dr. Mahmoud Khazalah Assistant Professor in Civil Engineering, Munib & Angela Masri Faculty of Engineering, Aqaba University of Technology, Jordan.	International Web Conference on Smart Engineering Technologies	17.06.2020	2 Hours	Invited Talk on Recent Trends in construction